If this paper is post on arXiv, I will post the link here: [] (updated, 12.1, 2025)

Complex-time Representation, Kime-Phase Tomography, and Spacekime Analytics (KPT V.4: Using the standard formulation of the time differentiation operator $\hat{P}_t = -i\hbar_\kappa \partial_t$)

Y. Shen,[1] Z. Tao,[2] B.N. Bakalov,[3] and I.D. Dinov[4, a)]
[1)] *Statistics Online Computational Resource, University of Michigan, Ann Arbor, MI*
[2)] *Statistics Online Computational Resource, University of Michigan, Ann Arbor, MI*
[3)] *Mathematics, North Carolina State University, Raleigh, NC*
[4)] *Statistics Online Computational Resource, University of Michigan, Ann Arbor, MI*

(*Electronic mail: statistics@umich.edu)

(Dated: 27 November 2025)

Abstract: We introduce *Kime-Phase Tomography* (KPT), a statistical-geometric framework for longitudinal data that augments chronological time $t \in [0,T]$ with a latent *phase* coordinate $\theta \in \mathbb{S}^1$ via the polar complex-time embedding $\kappa = te^{i\theta}$. The central objects are a *kime-surface* $\mathscr{S}(t,\theta)$, describing how expected responses vary across time and phase, and a *time-varying phase law* $\varphi_t(\theta)$ that quantifies structured, replicate-to-replicate variability. On each time slice the mixed moments obey a circular convolution $M = F \cdot \Phi$ between the surface harmonics F and the phase spectrum Φ, which we exploit for stable deconvolution. We place this model on the cone manifold $([0,T] \times \mathbb{S}^1, dt^2 + t^2 d\theta^2)$, construct the weighted Hilbert space $\mathscr{H} = L^2(t\,dt \otimes \frac{d\theta}{2\pi})$, and develop operator-theoretic tools (including a geometry-adjusted time generator $\tilde{P}_t$) that align the analysis with the manifold structure. We establish identifiability of φ_t from mixed moments under mild harmonic conditions, prove monotonicity and stationarity for a penalized generalized EM (GEM) scheme, and derive nonparametric consistency and finite-sample bounds for regularized estimators. A fast FFT-based implementation (KPT–FFT) yields Wiener–Sobolev updates per frequency with $\mathscr{O}(NKJ + KJ\log J)$ complexity for N replicates, K times, and J harmonic modes.

Empirically, we validate KPT on two contrasting settings. In a quantum double-slit simulation with drifting two-lobe phase structure, KPT disentangles latent phase variability from measurement noise and reconstructs the interference geometry; mapped through Fraunhofer optics, the recovered $(\widehat{\mathscr{S}}, \widehat{\varphi})$ produces calibrated predictive densities for screen positions, improving uncertainty quantification and enabling run-specific phase-offset inference. In a synthetic event-related fMRI study, KPT recovers time-varying concentration differences between ON and OFF blocks and reconstructs phase-modulated activation surfaces, offering interpretable summaries of run-to-run physiological state. Across both domains, KPT turns apparent noise into a low-dimensional geometric structure on $[0,T] \times \mathbb{S}^1$, supporting inference (Rayleigh- and V-type tests with de-biasing or bootstrap calibration), uncertainty bands via Fisher operators or bootstrap, and downstream prediction. Conceptually, imaginary time ($\theta = -\pi/2$) emerges as a distinguished kime slice; KPT leverages the full angular degree of freedom to model and infer latent, time-varying states in repeated measurements.

I. INTRODUCTION

Longitudinal data analysis underpins modern biomedical and physical sciences, enabling the study of disease progression, treatment effects, development, and dynamical behavior under repeated measurement. Standard approaches implicitly assume that variability across replicates at the same chronological time arises only from measurement noise and genuine temporal evolution. This overlooks an additional, often dominant, source of variation: *latent, time-varying phase states* that modulate the observed process independently of (and in addition to) classical longitudinal dynamics.

This paper introduces *spacekime analytics*, a framework that extends real-valued time $t \in \mathbb{R}_{\geq 0}$ to *complex time (kime)* $\kappa = te^{i\theta}$ with $t \geq 0$ and $\theta \in [0, 2\pi)$, thereby modeling both chronological evolution (the *amplitude*) and latent phase (the *argument*). The term "kime" distinguishes the complex-time domain from its amplitude $t = |\kappa|$ and makes explicit that repeated-measurement variability can be decomposed into (i) *domain-space* variability—intrinsic, time-varying phase structure—and (ii) *range-space* variability—extrinsic, typically additive measurement noise. In particular, the time axis $(\mathbb{R}_{\geq 0}, +)$ is totally ordered, whereas the kime domain $\mathbb{C}$ is not orderable as a field; moreover, $\mathbb{C}$ is the smallest algebraically closed field containing $\mathbb{R}$, a property that will enable spectral-analytic reconstructions on the phase circle. In practice, kime is the (polar) embedding $\kappa = te^{i\theta}$ of

[a)] https://SOCR.umich.edu/

$[0,T]\times\mathbb{S}^1\to\mathbb{C}$ and the induced *spacekime analytics* are performed on the product manifold with cone geometry, not on the complex field itself, see Section I B.

Complex-time relates to imaginary time reflecting the Wick rotation[1,2]. In quantum and statistical field theory, *Wick rotation* analytically continues real time to purely imaginary time in order to convert oscillatory unitary evolution into a decaying semigroup. Consider a self-adjoint Hamiltonian H describing the energy and a propagator $U(t)=\exp(-\frac{i}{\hbar}Ht)$, which tracks the system time evolution, where $\hbar$ is the reduced Planck constant. Then, the real-to-imaginary time mapping $t\mapsto -i\tau$ yields the Euclidean semigroup $U_E(\tau)=\exp(-\frac{1}{\hbar}\tau H)$, which underpins ground-state projection ($\tau\to\infty$), the Feynman-Kac formula, and finite-temperature partition functions $Z(\beta)=\mathrm{tr}(\exp(-\beta H))$ with periodic boundary conditions in imaginary time. This is justified by the analytic continuation of $U(t)$ into a narrow strip of the complex plane and (in axiomatic settings) by reflection positivity and reconstruction theorems. This continuation connects the Schrödinger equation with the heat equation, i.e., $i\hbar\partial_t\psi=H\psi$ becomes $\hbar\partial_\tau\psi=-H\psi$, where time and imaginary-time are connected by $t=-i\tau$. In statistical mechanics, the Gibbs measure $\exp(-\beta H)$ describes the relative probability of the system to be in any given state at temperature T, where k_B is the Boltzmann constant, and $\beta=1/(k_BT)$. In Euclidean (imaginary-time) formulations, finite temperature corresponds to periodicity in imaginary time of length β. Using the Wick rotation to replace $\frac{it}{\hbar}$ by $\frac{T}{k_B}$ is analogous to the quantum mechanics time evolution transformation. In Euclidean quantum field theory (QFT), finite-temperature corresponds to imaginary-time length β with periodic boundary conditions[3].

Also *kime* has a geometric interpretation through the polar representation of complex-time $\kappa=t\,e^{i\theta}$, where $t\ge 0$ and $\theta\in\mathbb{S}^1$ is the kime phase angle. From a semigroup perspective,

$$\exp\Big(-\tfrac{i}{\hbar}H\,\kappa\Big)=\exp\Big(-\tfrac{i}{\hbar}H\,t e^{i\theta}\Big)=\exp\Big(\tfrac{\sin\theta}{\hbar}\,t\,H\Big)\,\exp\Big(-\tfrac{i\cos\theta}{\hbar}\,t\,H\Big), \tag{1}$$

so each *kime ray* θ mixes a dissipative (Euclidean) factor and a unitary (real-time) factor. Note that since H commutes with itself, $[H,H]=0$, the last factorization in equ. (1) is exact from BCH formula. The classical imaginary-time via Wick rotation is the special slice $\theta=-\frac{\pi}{2}$, giving $U_E(t)=\exp(-\frac{t}{\hbar}H)$. Complex-time representation and the induced kime phase tomography (KPT), do not assert that measurement dynamics literally evolve along complex directions. Rather, we use the polar complex-time chart $\kappa=t\,e^{i\theta}$ to *separate* the chronological scale (t) from a latent *phase* coordinate (θ) that captures structured variability across repeated observations. The pair of KPT-recovered *kime-surface* $\mathscr{S}(t,\theta)$ and the *phase law* $\varphi_t(\theta)$ are analogous to a family of Euclideanized (repeated measurement longitudinal) responses along different complex directions and a distribution over those directions. We will demonstrate that this complex-time geometry makes the harmonic analysis in θ natural. The mixed moments satisfy the circular convolution $M=F\cdot\Phi$ on $\mathbb{S}^1$, enabling stable deconvolution (in both the KPT-GEM and KPT-FFT algorithms) to recover φ_t and $\mathscr{S}$ from replicated data. In this sense, *purely imaginary time is a distinguished kime slice*, while KPT leverages the full angular degree of freedom to model and infer latent phase structure in repeatedly observed longitudinal processes.

A. Driving Motivational Problems

To illustrate why repeated-measurement longitudinal data benefit from complex-time (kime) representation with an explicit phase dimension, consider the following motivational problems.

a. Quantum tunneling. In repeated tunneling trials near a potential barrier, arrival times vary even under matched initial conditions[4]. Rather than attributing all variation to randomness, we regard a latent, time-varying phase as modulating tunneling probability. Kime-phase tomography will reconstruct a surface $\mathscr{S}(t,\theta)$ whose θ-slices encode the latent kime-phase dependence of tunneling dynamics, turning apparent noise into a structured manifold over $\kappa=\kappa(t,\theta)$.

b. Turbulent flows. Velocity measurements downstream of an obstacle exhibit chaotic variation across runs[5]. Microscopic initial-condition sensitivity implies an internal, time-varying state; KPT deconvolves repeated velocity traces to estimate the phase distribution $\varphi_t(\theta)$ that summarizes the evolving state of the turbulent cascade.

c. Superconductor phase transitions. Repeated coolings across T_c yield variations in susceptibility trajectories[6]. These reflect unobserved microstructural configurations acting as a latent phase process. KPT reconstructs $\mathscr{S}(t,\theta)$ to expose phase-dependent transition pathways (fast/slow channels) that are otherwise obscured by averaging.

d. Neuroimaging. Run-to-run BOLD variability is usually modeled as noise, yet intrinsic biological cycles (e.g., circadian, metabolic) induce phase-dependent structure[7,8]. KPT separates genuine temporal effects from cross-sectional phase variability, improving reproducibility in repeated fMRI designs.

B. Mathematical Framework Overview

The central object is the *kime-surface* $\mathscr{S}:[0,T]\times\mathbb{S}^1\to\mathbb{R}$, with $\mathbb{S}^1=\{e^{i\theta}:\theta\in[0,2\pi)\}$ encoding intrinsic phase. We view the base space as $\mathscr{M}=[0,T]\times\mathbb{S}^1$ equipped with the cone (polar) metric $g_0=dt^2+t^2\,d\theta^2$ and volume element $d\mu_{g_0}=t\,dt\,d\theta$, corresponding to the embedding $\kappa = te^{i\theta}$. The graph of $\mathscr{S}$ in $\mathscr{M} \times \mathbb{R}$ induces a geometric structure that we regularize and compute with later (Section II).

For discrete observation times $\{t_k\}_{k=1}^{K}$, repeated measurements from run (or subject) j follow the additive model

$$y_{j,k} = \mathscr{S}(t_k, \Theta_j(t_k)) + \varepsilon_{j,k}, \tag{2}$$

where $\Theta_j(\cdot)$ is a *latent, time-varying phase process* on $\mathbb{S}^1$ (domain stochasticity) and $\varepsilon_{j,k}$ is additive measurement noise (range uncertainty). We assume $\mathbb{E}[\varepsilon_{j,k} \mid t_k] = 0$, $\mathrm{Var}(\varepsilon_{j,k} \mid t_k) < \infty$, and independence across j; unless otherwise stated, $\{\Theta_j(\cdot)\}$ and $\{\varepsilon_{j,k}\}$ are mutually independent. At each time t, the marginal phase density φ_t on $\mathbb{S}^1$ has Fourier coefficients

$$\hat{\varphi}_t(n) = \int_0^{2\pi} e^{-in\theta}\, \varphi_t(\theta)\, \frac{d\theta}{2\pi}, \qquad n \in \mathbb{Z},$$

a normalization consistent with the weighted L^2 geometry used later. The cone metric degeneracy at $t{=}0$ is handled by regularity at the apex or restriction to $[\varepsilon, T] \times \mathbb{S}^1$ (see Section II).

Section II formalizes the three pillars that support KPT:

a. Measure-Theoretic Foundations. We build probability on $(\mathscr{M}, \mathscr{B}, \mu_0)$ with μ_0 the cone volume and define the *kime Hilbert space* $\mathscr{H} = L^2(\mathscr{M}, \mu_0)$, enabling geometric penalties/stability aligned with the κ-plane.

b. Operator Theory and Tomography. On $\mathscr{H}$ we introduce non-commutative operators satisfying canonical commutation relations (standard time differentiation $\hat{P}_t = -i\hbar_\kappa \partial_t$), which enable spectral deconvolution on the circle for phase recovery.

c. Statistical Inference Theory. We state identifiability conditions for φ_t from mixed moments, derive efficient EM-type reconstructions with FFT acceleration, and establish nonparametric consistency and information-theoretic lower bounds.

C. Key Contributions

In this article, we present a probability theory on kime domains compatible with the cone geometry. We derive the *identifiability and consistency* results for latent phase distributions from repeated longitudinal data under verifiable conditions, Theorem 8, together with consistent nonparametric estimators Theorem 10 and we give nonparametric consistency and finite-sample bounds for φ_t and discuss parametric specialization, see Appendix VI. The kime representation supports *geometric statistical inference* on $[0, T] \times \mathbb{S}^1$, including tests for phase uniformity, confidence regions on the cylinder, and multiple testing adapted to circular domains. We propose *efficient KPT algorithms* and spectral numerical methods on $\mathbb{S}^1$ with complexity $\mathscr{O}(N \log N)$ in the number of phase modes, which are evaluated on a pair of experiments. The first example simulates a quantum double-slit experiment and the second example shows KPT performance on synthetic functional magnetic resonance imaging (fMRI) of stimulus (on) vs. rest (off) event-related experiments.

D. Related Work

We connect to phase retrieval[9,10] (deterministic phases) and circular statistics[11,12] (directional data) by extending to *time-varying* phase distributions on product manifolds. Work on functional data analysis[13,14] typically omits a latent phase dimension. The operator structure we develop also interfaces with emerging geometric deep learning[7,15]. Appendix VI situates the kime representation alongside Moyal's statistical formulation of quantum mechanics[16].

E. Paper Organization

Section II establishes probability, geometry, and operator theory on kime-space and introduces Bessel-Fourier decompositions. Section III develops Kime-Phase Tomography: identifiability of φ_t, penalized generalized EM with FFT implementations, and nonparametric theory. Section IV develops geometric inference on $[0, T] \times \mathbb{S}^1$. Section V presents simulations (double slit; event-related fMRI) and algorithmic evaluations. Section VI discusses implications and limitations. Finally, the Appendix VI contains supporting materials and a pointer to the complete computational notebook.

II. MATHEMATICAL FOUNDATIONS

This section establishes the rigorous mathematical framework for the complex-time (kime) representation of longitudinal processes. We develop the probability-theoretic foundations, the Riemannian geometric structure of kime-spaces, the operator theory enabling tomographic reconstruction, and the spectral decomposition that underpins efficient computational algorithms. We use the same notation introduced in Section I, which also directly supports the subsequent statistical theory in Section III.

A. Complex-Time (Kime) Representation

Definition 1 (Kime Domain)**.** The *kime domain* is the complex domain $\mathbb{K} = \{\kappa = te^{i\theta} : t \in \mathbb{R}_0^+,\ \theta \in [0, 2\pi)\}$, where $t \geq 0$ represents chronological time and $\theta \in [0, 2\pi)$ represents phase. For a finite observation window, we consider the restricted kime domain $\mathbb{K}_T = \{\kappa = te^{i\theta} : t \in [0, T], \theta \in [0, 2\pi)\}$.

The polar representation $\kappa = te^{i\theta}$ endows $\mathbb{K}_T$ with the structure of a product manifold $\mathscr{M} = [0, T] \times \mathbb{S}^1$, where $\mathbb{S}^1 = \{e^{i\theta} : \theta \in [0, 2\pi)\}$ is the unit circle. This structure is fundamental to the kime analytics framework, as it allows for the separate treatment of the radial (time) and angular (phase) coordinates.

Throughout the article we use the following notation: the *phase variable* $\theta \in \mathbb{S}^1$ with base measure $\frac{d\theta}{2\pi}$ is the Haar probability; *time*: $t \in [0, T]$ is measured over a discrete grid $\{t_k\}_{k=1}^K$; replicates (subjects/runs) are indexed by $j = 1, \ldots, N$; the *harmonic index* $n \in \{-J, \ldots, J\}$; the DFT frequencies are indexed by $\omega = \frac{2\pi k}{2J+1}$ for $k = -J, \ldots, J$; the *kime-surface* Fourier series is $\mathscr{S}(t, \theta) = \sum_n f_n(t) e^{in\theta}$ with $f_{-n} = \overline{f_n}$; the *phase density* is $\varphi_t(\theta) \geq 0$ where $\int \varphi_t(\theta)\, \frac{d\theta}{2\pi} = 1$; and the Fourier coefficients are defined by $\hat{\varphi}_t(n) = \int e^{-in\theta} \varphi_t(\theta)\, \frac{d\theta}{2\pi}$, with $\hat{\varphi}_t(-n) = \overline{\hat{\varphi}_t(n)}$.

Definition 2 (Kime Process)**.** A *kime process* is a measurable function $X : \Omega \times \mathscr{M} \to \mathbb{R}$ defined on a probability space $(\Omega, \mathscr{F}, \mathbb{P})$. For a fixed outcome $\omega \in \Omega$, the function $X(\omega, \cdot) : \mathscr{M} \to \mathbb{R}$ is called a sample path.

The key distinction from a classical stochastic process $X(t)$ is the additional phase dimension. While a classical process models variation as additive noise in the range space, a kime process $X(t, \theta)$ explicitly models intrinsic, time-varying variability in the domain space via the latent stochastic process $\Theta(t)$.

B. Probability Spaces and Measure Theory

We consider kime processes on the measurable space $(\mathscr{M}, \mathscr{B}(\mathscr{M}))$, where $\mathscr{B}(\mathscr{M}) = \mathscr{B}([0, T]) \otimes \mathscr{B}(S^1)$ is the product Borel σ-algebra.

Definition 3 (Time-Varying Phase Measure)**.** A *time-varying phase measure* is a Markov kernel $t \mapsto \mu_{\theta,t}$ on $(\mathbb{S}^1, \mathscr{B}(\mathbb{S}^1))$ such that for each $t \in [0, T]$, $\mu_{\theta,t}$ is a probability measure, and for each Borel set $B \in \mathscr{B}(\mathbb{S}^1)$, the function $t \mapsto \mu_{\theta,t}(B)$ is $\mathscr{B}([0, T])$-measurable. If $\mu_{\theta,t}$ is absolutely continuous with respect to the Lebesgue (Haar) measure on $\mathbb{S}^1$, its *kime-phase density* is denoted $\varphi_t(\theta) \geq 0$, normalized such that $\int_0^{2\pi} \varphi_t(\theta) \frac{d\theta}{2\pi} = 1$.

Proposition 1 (Existence of Regular Conditional Distributions)**.** *Fix $t \in [0, T]$. Let $X : \Omega \times \mathscr{M} \to \mathbb{R}$ be a kime process with finite second moments, and let $\Theta(t) : \Omega \to \mathbb{S}^1$ be a measurable phase variable with density φ_t. Then there exists a regular conditional distribution $\mathscr{K}(\cdot; t, \theta)$ on $(\mathbb{R}, \mathscr{B}(\mathbb{R}))$ such that for every Borel set $B \subset \mathbb{R}$,*

$$\mathbb{P}\big(X(t, \Theta(t)) \in B\big) = \int_0^{2\pi} \mathscr{K}(B; t, \theta) \varphi_t(\theta) \frac{d\theta}{2\pi},$$

where the map $(t, \theta) \mapsto \mathscr{K}(B; t, \theta)$ is measurable.

Proof. For each fixed t, the spaces $(\mathbb{R}, \mathscr{B}(\mathbb{R}))$ and $(\mathbb{S}^1, \mathscr{B}(\mathbb{S}^1))$ are standard Borel spaces. By the Disintegration Theorem[17] (Thm. 6.4), a regular version of the conditional probability $\mathbb{P}(X(t, \cdot) \in \cdot \mid \Theta(t) = \theta)$ exists. Integrating this against the marginal density $\varphi_t(\theta) \frac{d\theta}{2\pi}$ yields the stated identity. □

This proposition provides the measure-theoretic foundation for the mixed-moment identities and the identifiability theorem used later.

C. Kime-Surface Geometry

Definition 4 (Base Manifold Structure)**.** The *base manifold* is the Riemannian manifold $(\mathscr{M}, g_0)$, where $\mathscr{M} = [0, T] \times \mathbb{S}^1$ and g_0 is the *polar metric* given by $g_0 = dt^2 + t^2 d\theta^2$. The associated *volume element* is $d\mu_{g_0} = \sqrt{\det g_0} dt d\theta = t\, dt\, d\theta$.

A key quantity for regularization is the squared norm of the surface gradient with respect to this cone metric, which defines the *cone-Dirichlet energy*

$$\|\nabla_{g_0}\mathscr{S}\|^2 = (\partial_t\mathscr{S})^2 + \frac{1}{t^2}(\partial_\theta\mathscr{S})^2.$$

Definition 5 (Kime-Surface)**.** A *kime-surface* is a measurable function $\mathscr{S}:\mathscr{M}\to\mathbb{R}$ satisfying: (i) $\mathscr{S}\in L^2(\mathscr{M},d\mu_{g_0})$; (ii) $\mathscr{S}(t,\theta+2\pi)=\mathscr{S}(t,\theta)$ for all $t\in[0,T]$; (iii) For almost every $t\in[0,T]$, the Fourier series $\mathscr{S}(t,\theta)=\sum_{n\in\mathbb{Z}} f_n(t)e^{in\theta}$ converges in $L^2(\mathbb{S}^1)$.

Proposition 2 (Induced Metric)**.** *Let $\mathscr{S}\in C^2(\mathscr{M})$ and define the embedding $F:\mathscr{M}\to\mathbb{R}^3$ by*

$$F(t,\theta) = (t\cos\theta, t\sin\theta, \mathscr{S}(t,\theta)). \tag{3}$$

The Riemannian metric g induced on $\mathscr{M}$ by this embedding is given by the matrix

$$g = \begin{pmatrix} 1+(\partial_t\mathscr{S})^2 & (\partial_t\mathscr{S})(\partial_\theta\mathscr{S}) \\ (\partial_t\mathscr{S})(\partial_\theta\mathscr{S}) & t^2+(\partial_\theta\mathscr{S})^2 \end{pmatrix}. \tag{4}$$

Proof. The partial derivatives of the embedding are $F_t=(\cos\theta,\sin\theta,\partial_t\mathscr{S})$ and $F_\theta=(-t\sin\theta,t\cos\theta,\partial_\theta\mathscr{S})$. The components of the induced metric are the inner products $g_{ij}=\langle F_i,F_j\rangle_{\mathbb{R}^3}$, which directly yield the matrix entries. □

D. Operator Theory Framework

Let us define the principal kime operators defined over the kime Hilbert space of functions.

Definition 6 (Kime Hilbert Space)**.** The *kime Hilbert space* is the space of square-integrable functions on $\mathscr{M}$ with respect to the cone volume measure

$$\mathscr{H} = L^2\left(\mathscr{M}, t\,dt\otimes\frac{d\theta}{2\pi}\right),$$

equipped with the inner product

$$\langle\psi,\phi\rangle = \int_0^T\int_0^{2\pi}\psi(t,\theta)\overline{\phi(t,\theta)}t\frac{d\theta}{2\pi}dt.$$

Definition 7 (Fundamental Kime Operators)**.** On the dense space of smooth kime functions $\mathscr{D}\subset\mathscr{H}$, we define:

$$\begin{aligned}
(\hat{T}\psi)(t,\theta) &= t\psi(t,\theta), && \text{(time multiplication operator)}\\
(\hat{P}_t\psi)(t,\theta) &= -i\hbar_\kappa\partial_t\psi(t,\theta), && \text{(time differentiation operator)}\\
(\hat{\Theta}\psi)(t,\theta) &= e^{i\theta}\psi(t,\theta), && \text{(phase multiplication operator)}\\
(\hat{L}_\theta\psi)(t,\theta) &= -i\hbar_\kappa\partial_\theta\psi(t,\theta), && \text{(phase differentiation operator)}
\end{aligned}$$

where $\hbar_\kappa>0$ is the *kime-Planck constant*.

Proposition 3 (Canonical Commutation Relations)**.** *On the space of kime functions $\mathscr{D}$, the fundamental kime operators satisfy*

$$[\hat{T},\hat{P}_t] = i\hbar_\kappa\mathbf{I}, \qquad [\hat{L}_\theta,\hat{\Theta}] = \hbar_\kappa\hat{\Theta}.$$

Proof. For any $\psi\in\mathscr{D}$, the time commutator is

$$[\hat{T},\hat{P}_t]\psi = t(-i\hbar_\kappa\partial_t\psi) - (-i\hbar_\kappa)\partial_t(t\psi) = -i\hbar_\kappa t\psi_t + i\hbar_\kappa(t\psi_t+\psi) = i\hbar_\kappa\psi.$$

For the phase operators, let $\psi(t,\theta)=\sum_m c_m(t)e^{im\theta}$. The operator $\hat{\Theta}$ acts as a raising operator on the Fourier modes ($m\mapsto m+1$). We have $\hat{L}_\theta\psi=\sum_m\hbar_\kappa m c_m(t)e^{im\theta}$. Then,

$$(\hat{L}_\theta\hat{\Theta}-\hat{\Theta}\hat{L}_\theta)\psi = \sum_m c_m(t)(\hbar_\kappa(m+1)-\hbar_\kappa m)e^{i(m+1)\theta} = \hbar_\kappa\sum_m c_m(t)e^{i(m+1)\theta} = \hbar_\kappa\hat{\Theta}\psi.$$

This proves the Weyl-type relation $[\hat{L}_\theta,\hat{\Theta}]=\hbar_\kappa\hat{\Theta}$. □

Critical Note on the Symmetry of $\hat{P}_t$: The operator $\hat{P}_t = -i\hbar_\kappa \partial_t$ is not symmetric (Hermitian) in the weighted Hilbert space $\mathscr{H}$, as integration by parts picks up an extra term from the measure. The corresponding *symmetric*, or *geometry-adjusted*, operator is

$$\tilde{P}_t = -i\hbar_\kappa \left(\partial_t + \frac{1}{2t}\right).$$

This operator $\tilde{P}_t$ is unitarily equivalent to the standard operator $\hat{P}_t$ on the *unweighted* space $L^2(\mathscr{M}, dt \otimes \frac{d\theta}{2\pi})$ via the map $(U\psi)(t,\theta) = \sqrt{t}\,\psi(t,\theta)$. Crucially, both operators satisfy the same algebraic commutation relation, $[\hat{T}, \cdot] = i\hbar_\kappa \mathbf{I}$. For clarity in algebraic derivations, we use $\hat{P}_t$, but for spectral theory, symmetry is properly understood through $\tilde{P}_t$. We take $\mathscr{D} = C_c^\infty((0,T)\times\mathbb{S}^1)$ as a core space of kime functions. $\tilde{P}_t$ admits self-adjoint extensions compatible with regularity at $t = 0$ (Bessel behavior) or with a small apex excision $[\varepsilon,\ T]$. At the singularity $t = 0$, we either impose regularity conditions consistent with the Bessel behavior of the radial part (see Section VI), or alternatively restrict the kime function space to $[\varepsilon, T]\times\mathbb{S}^1$ and let $\varepsilon \downarrow 0$.

Proposition 4 (Operator Properties)**.** *On the space of kime functions $\mathscr{D}$:*

- *The time multiplication operator $\hat{T}$ is a bounded and symmetric operator.*
- *The phase multiplication operator $\hat{\Theta}$ is a unitary operator.*
- *The phase differentiation operator $\hat{L}_\theta$ is essentially self-adjoint on the space of smooth, 2π-periodic functions.*
- *The geometry-adjusted time differentiation operator $\tilde{P}_t = -i\hbar_\kappa(\partial_t + \frac{1}{2t})$ is symmetric on smooth functions with compact support in $(0,T)$ and admits self-adjoint extensions.*

E. Spectral Theory and Basis Functions

The *Bessel-Fourier basis*, see Appendix VI, provides a geometrically natural dictionary for kime-surfaces by diagonalizing the Laplace-Beltrami operator on the kime manifold.

Definition 8 (Laplace-Beltrami Operator)**.** On the kime manifold $(\mathscr{M}, g_0)$, the Laplace-Beltrami operator Δ is

$$\Delta_{g_0} = \frac{1}{t}\frac{\partial}{\partial t}\left(t\frac{\partial}{\partial t}\right) + \frac{1}{t^2}\frac{\partial^2}{\partial\theta^2}.$$

Theorem 5 (Spectral Decomposition)**.** *The operator $-\Delta_{g_0}$ on $(0,T)\times\mathbb{S}^1$ with a regularity condition at $t = 0$ and a Dirichlet boundary condition at $t = T$ has a complete orthonormal basis of eigenfunctions $\{\phi_{n,m}\}$ in $\mathscr{H}$ of the form*

$$\phi_{n,m}(t,\theta) = R_{n,m}(t)e^{im\theta}, \qquad -\Delta_{g_0}\phi_{n,m} = \mu_{n,m}\phi_{n,m},$$

where $R_{n,m}(t) = A_{n,m}J_{|m|}(j_{|m|,n}t/T)$ and $\mu_{n,m} = (j_{|m|,n}/T)^2$, with $j_{|m|,n}$ being the n-th positive zero of the Bessel function $J_{|m|}$. Completeness follows from Sturm-Liouville theory on the weighted space $([0,T], t\,dt)$[18,19].

F. Fourier Analysis on Kime-Spaces

Definition 9 (Kime Fourier Transform)**.** For a kime surface $\mathscr{S} \in L^2(\mathscr{M}, d\mu_{g_0})$, the phase-Fourier coefficients are

$$f_n(t) = \int_0^{2\pi} \mathscr{S}(t,\theta)e^{-in\theta}\frac{d\theta}{2\pi}, \qquad n \in \mathbb{Z}.$$

For almost every (a.e.) t, we have $\mathscr{S}(t,\theta) = \sum_{n\in\mathbb{Z}} f_n(t)e^{in\theta}$ in $L^2(\mathbb{S}^1)$.

Proposition 6 (Parseval's Identity with Cone Weight)**.** *For any $\mathscr{S} \in L^2(\mathscr{M}, d\mu_{g_0})$, we have*

$$\int_0^T\int_0^{2\pi} |\mathscr{S}(t,\theta)|^2 t\,dt\,\frac{d\theta}{2\pi} = \int_0^T \sum_{n\in\mathbb{Z}} |f_n(t)|^2 t\,dt.$$

The Sobolev regularization penalties used in the KPT algorithm are implemented efficiently in this Fourier domain. The continuous penalty on the phase derivative,

$$\|\partial_\theta^p \varphi\|^2_{L^2(\mathbb{S}^1)} = \sum_{n\in\mathbb{Z}} n^{2p}\,|\hat{\varphi}(n)|^2,$$

has a discrete counterpart on a grid that is diagonal in the Discrete Fourier Transform (DFT) domain, with weights $\Lambda_p(\omega) = (2\sin(\frac{\omega}{2}))^{2p}$. This is the exact regularizer used in the M-step (optimization) of the KPT-FFT algorithm.

G. Information Geometry

Definition 10 (Fisher Information Metric)**.** Let $\{\varphi_\alpha : \mathbb{S}^1 \to (0,\infty)\}_{\alpha\in\mathscr{A}\subset\mathbb{R}^d}$ be a smooth, parametric family of phase densities. The *Fisher information metric* on the parameter manifold $\mathscr{A}$ at α is:

$$g^F_{ij}(\alpha) = \int_0^{2\pi} \frac{1}{\varphi_\alpha(\theta)} \frac{\partial \varphi_\alpha(\theta)}{\partial \alpha_i} \frac{\partial \varphi_\alpha(\theta)}{\partial \alpha_j} \frac{d\theta}{2\pi}.$$

Note that the Fisher metric integrates over θ, the sample space, not the parameter manifold. As the integration is over the circle, $\mathbb{S}^1$, the support of the observation, the Fisher metric resides on the parameter manifold $\mathscr{A}$.

Example (Information-Geometric Structure of the von Mises Family): Consider the von Mises family with a (location, concentration) parameter vector $\alpha = (\mu,\kappa)$. Then, the density is $\varphi(\theta;\mu,\kappa) = \left(2\pi I_0(\kappa)\right)^{-1} \exp\{\kappa\cos(\theta-\mu)\}$, where $I_j(x)$ is the modified Bessel function of order j[20], and the score functions are

$$\partial_\mu \log\varphi = -\kappa\sin(\theta-\mu), \qquad \partial_\kappa \log\varphi = \cos(\theta-\mu) - \frac{I_1(\kappa)}{I_0(\kappa)}.$$

The Fisher information metric is a diagonal square matrix

$$g^F(\mu,\kappa) = \begin{pmatrix} \frac{\kappa^2}{2}\left(1-A_2(\kappa)\right) & 0 \\ 0 & \frac{1}{2}\left(1+A_2(\kappa)\right) - A_1(\kappa)^2 \end{pmatrix},$$

where $A_m(\kappa) = I_m(\kappa)/I_0(\kappa)$.

This metric provides a coordinate-invariant geometry for the space of phase distributions, which is foundational for developing robust statistical inference procedures.

III. KIME-PHASE TOMOGRAPHY (KPT)

This section develops the statistical theory for recovering time-varying phase distributions from longitudinal observations. We establish identifiability, present tomographic reconstruction procedures, and prove convergence and risk guarantees under assumptions aligned with the geometric/operational framework of Section II, cone metric, phase-Fourier conventions with $d\theta/(2\pi)$, and a unified kime-Planck scale $\hbar_\kappa$.

A. Kime Generating Functions

Let $\mathscr{S}(t,\theta) = \sum_{n\in\mathbb{Z}} f_n(t)e^{in\theta}$ with $f_{-n}(t) = \overline{f_n(t)}$ so that the kime-surface $\mathscr{S}$ is real-valued. We define the following three generating functions on the unit circle $|z| = 1$:

- *Kime-surface transform* $F(z,t) = \sum_{n\in\mathbb{Z}} f_n(t)\,z^n$;
- *Phase transform* $\Phi(z,t) = \sum_{n\in\mathbb{Z}} \hat{\varphi}_t(n)\,z^n$, with $\hat{\varphi}_t(n) = \int e^{-in\theta}\varphi_t(\theta)\,\frac{d\theta}{2\pi}$ and $\hat{\varphi}_t(0) = 1$;
- *Mixed-moment transform* $M(z,t) = \sum_{n\in\mathbb{Z}} m_n(t)\,z^n$, where $m_n(t) = \mathbb{E}[\mathscr{S}(t,\Theta(t))e^{-in\Theta(t)}]$.

As in Section II, we use the $d\theta/(2\pi)$ normalization, so that Parseval and FFT implementations match exactly. *All* spectral computations below use the discrete grid $\omega = \frac{2\pi k}{2J+1}$ for $k = -J,\ldots,J$, consistent with the $(2J+1)$-point DFT used in code, i.e., the KPT algorithmic implementation.

Definition 11 (Mixed Moments)**.** Given a kime-surface $\mathscr{S}$ and a time-varying phase distribution φ_t, as described above, the *mixed moments* are

$$m_n(t) = \sum_{k\in\mathbb{Z}} f_k(t)\,\hat{\varphi}_t(n-k), \qquad n\in\mathbb{Z}.$$

We assume absolute convergence of the mixed-moment transform M and the Fourier coefficients of $\mathscr{S}(\cdot,t)$ are absolutely summable for a.e. t, i.e., $\sum_n |f_n(t)| < \infty$ and $\sum_n |\hat{\varphi}_n(t)| < \infty$.

Lemma 7 (Convolution Structure)**.** *For each fixed t and all $|z| = 1$, we have*

$$M(z,t) = F(z,t)\,\Phi(z,t).$$

Proof. This is circular convolution in the phase-harmonic index. It follows by direct rearrangement of the absolutely summable series under Definition 11. □

Global phase-shift ambiguity and anchoring. We fix the rotation by imposing $\arg\widehat{\varphi}_t(1) = 0$ (or, equivalently, $\Re f_1(t) \geq 0$), so that $\widehat{\varphi}_t$ and $\widehat{\mathscr{S}}$ are jointly anchored. If we rotate the phase coordinate by $\alpha(t)$, then $F(z,t) \mapsto F(ze^{i\alpha(t)},t)$ and $\Phi(z,t) \mapsto \Phi(ze^{-i\alpha(t)},t)$ leaving M unchanged. We therefore identify φ_t *up to a circular shift* unless an *anchoring convention* is imposed, for instance by forcing $\arg\hat{\varphi}_t(1) = 0$ or $\Re f_1(t) \geq 0$. This convention is used in all KPT reconstructions.

B. Identifiability Theory

Theorem 8 (Identifiability of Phase Distributions)**.** *Fix $t \in (0,T]$ and suppose the following:*

- *(I1)* ***Coprimality of the set of active harmonics of*** *F: $\gcd\{n \in \mathbb{Z} : f_n(t) \neq 0\} = 1$. This rules out aliasing by a common divisor in the active harmonic set.*
- *(I2)* ***Non-vanishing on $\mathbb{S}^1$ (exact recovery):*** *$F(\cdot,t)$ is continuous on $|z| = 1$ and $\inf_{|z|=1}|F(z,t)| > 0$, i.e., non-vanishing F on the unit circle, for exact recovery, or Tikhonov regularization for stable deconvolution if F has zeros;*
- *(I3)* ***Regularity:*** *$\varphi_t \in L^1(\mathbb{S}^1)$ with $\hat{\varphi}_t \in \ell^1(\mathbb{Z})$, $\hat{\varphi}_t(-n) = \overline{\hat{\varphi}_t(n)}$, and $\hat{\varphi}_t(0) = 1$, i.e., anchoring.*

Then the map $\varphi_t \mapsto \{m_n(t)\}_{n\in\mathbb{Z}}$ is injective (up to a circular shift). In particular, if two densities yield the same mixed moments, their phase transforms coincide on $|z| = 1$ and hence agree in $L^1(\mathbb{S}^1)$.

Proof. By Lemma 7, assuming the equality $F\,\Phi^{(1)} = F\,\Phi^{(2)}$ and non-vanishing of F on $|z| = 1$, we have equality of $\Phi^{(1)} = \Phi^{(2)}$ on $|z| = 1$, and hence equality of $\hat{\varphi}_t^1(n) = \hat{\varphi}_t^2(n)$ by Fourier inversion. Assumption (I2) implies $\Phi^{(1)} = \Phi^{(2)}$; uniform convergence of the absolutely summable Fourier series then gives equality of coefficients. Assumption (I1) rules out alias-type degeneracies (subsampling by a common divisor). The global phase-shift ambiguity is removed by the chosen anchoring. In other words, any global shift ambiguity is resolved by the anchoring convention $\arg\varphi_t(1) = 0$, see the see global phase-shift ambiguity and anchoring remark earlier. □

Stable deconvolution (practical identifiability). If $F(\cdot,t)$ has isolated zeros on $|z| = 1$, exact invertibility fails, but the Tikhonov regularizer used below yields a unique *regularized* solution, and modes at or near zeros are recoverable up to the bias induced by the penalty. This is precisely the setting of the FFT-based Wiener-Sobolev filter used in the KPT *M-step*.

C. Tomographic Reconstruction via Alternating Minimization

We reconstruct the phase distribution and the kime-surface, $(\varphi_t, \mathscr{S})$, by alternating between phase and surface updates. Let $\Lambda_p(\omega) = (2\sin(\frac{\omega}{2}))^{2p}$ be the discrete Sobolev weight of order $p \in \{0,1,2\}$ and let $\Pi_{\mathscr{P}}$ denote Euclidean projection of $\widehat{\varphi}_{t_k}$ onto the probability simplex $\{\phi(\theta) \geq 0,\ \int \phi(\theta)\,\frac{d\theta}{2\pi} = 1\}$.

Posterior moments (E-step). For noise $Y \mid \theta \sim \mathsf{N}(\mathscr{S}(t,\theta),\sigma^2)$, we have

$$p(\theta \mid y;t) \;\propto\; \varphi_t(\theta)\exp\!\left(-\frac{(y-\mathscr{S}(t,\theta))^2}{2\sigma^2}\right), \qquad \xi_{j,k}(n) = \int_0^{2\pi} e^{-in\theta}\,p(\theta \mid y_{j,k};t_k)\,\frac{d\theta}{2\pi}.$$

Empirical mixed moments at t_k are $\widehat{m}_n(t_k) = \frac{1}{N}\sum_{j=1}^{N} y_{j,k}\,\xi_{j,k}(n)$ and we assume that the recovered kime-surface approximates the observed repeated measurements, i.e., $y_{j,k} \approx \mathscr{S}(t_k,\theta_{j,k})$. This leads to obtaining sub-Gaussian alternatives for $p(y \mid \theta)$. In practice, to align the theoretical framework, analytical representation, and algorithmic code, all θ-integrals in $\xi_{j,k}(n)$ can be computed by FFT quadrature on the $(2J+1)$-point grid.

Note that for each frequency-time pair solves a strictly convex quadratic optimization problem whenever $|\widehat{F}|^2 + \lambda_\Phi\Lambda_{p_\Phi} > 0$ (analogously for the surface update in the F-step). When $p \geq 1$, adding a tiny ridge at $\omega = 0$ guarantees strict positivity. Also, the projection $\Pi_{\mathscr{P}}$ is the Euclidean projection onto the probability simplex under $d\theta/(2\pi)$ normalization, which enforces nonnegativity and unit mass. When working or calibrating with a known kime-surface, the mode sets $\lambda_F = \infty$, which suggests skipping the F-step in the KPT algorithm.

Algorithm 1: Kime-Phase Tomography generalized EM algorithm (KPT-GEM)

1: **Inputs:** $\{y_{j,k}\}$, J, harmonics $n \in \{-J,\dots,J\}$, tolerance ε, (λ_Φ, p_Φ), (λ_F, p_F)

2: **Initialize:** $\varphi_{t_k}^{(0)} \equiv \frac{1}{2\pi}$; initialize $f_n^{(0)}(t_k)$ (e.g., from cross-sectional averages); set iteration $\ell = 0$

3: **repeat**

4: **E-step:** Compute posterior expectations

$$\xi_{j,k}^{(\ell)}(n) = \mathbb{E}[e^{-in\Theta_{j,k}}|y_{j,k}] = \int_0^{2\pi} e^{-in\theta} w_{j,k}^{(\ell)}(\theta)\frac{d\theta}{2\pi}$$

where $w_{j,k}^{(\ell)}(\theta) \propto \varphi_{t_k}^{(\ell-1)}(\theta)\exp\left(-\frac{(y_{j,k}-\mathscr{S}^{(\ell-1)}(t_k,\theta))^2}{2\sigma^2}\right)$, and the mixed moments

$$m_n^{(\ell)}(t_k) = \tfrac{1}{N}\sum_j y_{j,k}\xi_{j,k}^{(\ell)}(n)$$

(Below, the **M-step** involves optimization of Φ, phase, and F, kime-surface)

5: **Φ-step (phase update):** For each t_k, FFT in n to get $M^{(\ell)}(\omega,t_k)$ and $F^{(\ell)}(\omega,t_k)$; set

$$\Phi^{(\ell+1)}(\omega,t_k) = \frac{\overline{F^{(\ell)}(\omega,t_k)}}{|F^{(\ell)}(\omega,t_k)|^2 + \lambda_\Phi \Lambda_{p_\Phi}(\omega)} M^{(\ell)}(\omega,t_k).$$

IFFT & projection: $\varphi_{t_k}^{(\ell+1)} \leftarrow \Pi_{\mathscr{P}}\left[\mathscr{F}^{-1}\Phi^{(\ell+1)}(\cdot,t_k)\right]$. Optionally smooth in t.

6: **Optional F-step (surface update):** For each t_k, recompute $\Phi^{(\ell+1)}$ and set

$$F^{(\ell+1)}(\omega,t_k) = \frac{\overline{\Phi^{(\ell+1)}(\omega,t_k)}}{|\Phi^{(\ell+1)}(\omega,t_k)|^2 + \lambda_F \Lambda_{p_F}(\omega)} M^{(\ell)}(\omega,t_k).$$

7: **IFFT & reality:** $f_n^{(\ell+1)}(t_k) \leftarrow \mathrm{IFFT}_\omega\{F^{(\ell+1)}(\omega,t_k)\}$ and enforce $f_{-n} = \overline{f_n}$

8: **Synthesis:** $\mathscr{S}^{(\ell+1)}(t_k,\theta) = \sum_{n=-J}^{J} f_n^{(\ell+1)}(t_k)e^{in\theta}$

9: **Convergence check:** stop when $\max_k \|\varphi_{t_k}^{(\ell+1)} - \varphi_{t_k}^{(\ell)}\|_{L^2} < \varepsilon$ and $\max_{k,n}|f_n^{(\ell+1)}(t_k) - f_n^{(\ell)}(t_k)| < \varepsilon$

10: **until** converged

11: **Return:** $\varphi_t^{(L)}$ and $\mathscr{S}^{(L)}(t,\theta)$

Algorithm 2: Fast Kime-Phase Tomography Algorithm via FFT (KPT-FFT)

1: **Input:** $\{y_{j,k}\}$, J, harmonics $n \in \{-J,\dots,J\}$, tolerance ε, (λ_Φ, p_Φ), (λ_F, p_F)

2: **Initialize:** $\varphi_{t_k}^{(0)} \equiv \frac{1}{2\pi}$; initialize $f_n^{(0)}(t_k)$

3: **repeat**

4: **E-step (vectorized):** Compute posterior expectations

$$\xi_{j,k}^{(\ell)}(n) = \mathbb{E}[e^{-in\Theta_j}|y_{j,k}] = \int_0^{2\pi} e^{-in\theta} w_{j,k}^{(\ell)}(\theta)\frac{d\theta}{2\pi}$$

where $w_{j,k}^{(\ell)}(\theta) \propto \varphi_{t_k}^{(\ell-1)}(\theta)\exp\left(-\frac{(y_{j,k}-\mathscr{S}^{(\ell-1)}(t_k,\theta))^2}{2\sigma^2}\right)$, and the mixed moments

$$m_n^{(\ell)}(t_k) = \tfrac{1}{N}\sum_j y_{j,k}\xi_{j,k}^{(\ell)}(n)$$

(Again, the **M-step** involves optimization of Φ, phase, and F, kime-surface)

5: **Φ-step (phase update):** For each t_k, FFT in n (the DFT grid uses the harmonic grid and ridge at $\omega = 0$ when $p \geq 1$) to get $M^{(\ell)}(\omega,t_k)$ and $F^{(\ell)}(\omega,t_k)$;

$$\Phi^{(\ell+1)}(\omega,t_k) \leftarrow \frac{\overline{F^{(\ell)}(\omega,t_k)}}{|F^{(\ell)}(\omega,t_k)|^2 + \lambda_\Phi \Lambda_{p_\Phi}(\omega)} M^{(\ell)}(\omega,t_k)$$

6: **IFFT & projection:** $\varphi_{t_k}^{(\ell+1)} \leftarrow \Pi_{\mathscr{P}}[\mathrm{IFFT}_\omega\{\Phi^{(\ell+1)}(\omega,t_k)\}]$

7: **F-step (surface update):** For each t_k, recompute $\Phi^{(\ell+1)}$ and set

$$F^{(\ell+1)}(\omega,t_k) \leftarrow \frac{\overline{\Phi^{(\ell+1)}(\omega,t_k)}}{|\Phi^{(\ell+1)}(\omega,t_k)|^2 + \lambda_F \Lambda_{p_F}(\omega)} M^{(\ell)}(\omega,t_k)$$

8: **IFFT & reality:** $f_n^{(\ell+1)}(t_k) \leftarrow \mathrm{IFFT}_\omega\{F^{(\ell+1)}(\omega,t_k)\}$ and enforce $f_{-n} = \overline{f_n}$

9: **Synthesis:** $\mathscr{S}^{(\ell+1)}(t_k,\theta) = \sum_{n=-J}^{J} f_n^{(\ell+1)}(t_k)e^{in\theta}$

10: **Convergence check:** stop when $\max_k \|\varphi_{t_k}^{(\ell+1)} - \varphi_{t_k}^{(\ell)}\|_{L^2} < \varepsilon$ and $\max_{k,n}|f_n^{(\ell+1)}(t_k) - f_n^{(\ell)}(t_k)| < \varepsilon$

11: **until** converged

12: **Return:** $\varphi_t^{(L)}$ and $\mathscr{S}^{(L)}(t,\theta)$

Notes: (1) The iteration symbol $\varphi_{t_k}^{(\ell)}$ natively indicates it being an estimated quantity. We remove the $\widehat{\cdot}$ and use capital letters (Φ, F, M) to indicate estimates, to avoid conflict with the notation of a Fourier-transformed quantity.

(2) KPT-FFT is the same as KPT-GEM but always alternates between Φ-step and F-step updates at each iteration. The Sobolev penalty $\Lambda_p(\omega) = (2\sin(\omega/2))^{2p}$ controls smoothness.

Theorem 9 (Monotonicity and Stationarity of Penalized GEM)**.** *Since per-frequency M-steps are strictly convex due to $|\hat{F}|^2 + \lambda\Lambda > 0$, and assuming an exact E-step, let*

$$\mathscr{L}_\lambda(\varphi;\widehat{f}) = \sum_{j,k}\left(\log\int p(y_{j,k} \mid \theta;\widehat{f})\,\varphi_{t_k}(\theta)\,\tfrac{d\theta}{2\pi}\right) - \lambda_\Phi\|\partial_\theta^{p_\Phi}\varphi\|_{L^2}^2. \tag{5}$$

Using conditions (I1)–(I3) in Theorem 8 and assuming σ^2 is known (or can be consistently estimated), the iteration in Algorithm 1 produces a non-decreasing sequence $\mathscr{L}_\lambda(\varphi^{(\ell)}; f^{(\ell)})$ where every accumulation point is a stationary point of the penalized objective. The per-frequency subproblems are strictly convex under the positivity conditions above.

The penalized KPT-GEM algorithm assumes the likelihood $p(y \mid \theta; f^{(L)})$ is correctly specified, e.g., Gaussian with known σ^2 and the E-step uses the exact posterior moments. The proof of Theorem 9 relies on the Jensen's inequality for the complete-data surrogate and the convex quadratic M-steps.

D. KPT Consistency and Finite-Sample Guarantees

Theorem 10 (Consistency of Non-parametric Phase Estimation)**.** *Under (I1)-(I3), stationarity/ergodicity of* $\{\Theta_j(t)\}$*,* $\varepsilon_{j,k} \sim \mathsf{N}(0,\sigma^2)$*, dense* $\{t_k\}$ *in* $[0,T]$*, and* $\lambda_{\Phi,N} = O(N^{-1/2})$*, for fixed* t*, we have*

$$\|\widehat{\varphi}_{t,N} - \varphi_t\|_{L^2(\mathbb{S}^1)} \xrightarrow{P} 0, \qquad \|\widehat{\varphi}_{t,N} - \varphi_t\|_{L^2(\mathbb{S}^1)} = O_P(N^{-1/4}).$$

Proof Sketch. Bias-variance decomposition: $\|\widehat{\varphi}_{t,N} - \varphi_t\| \le \|\widehat{\varphi}_{t,N} - \varphi_{t,\lambda}\| + \|\varphi_{t,\lambda} - \varphi_t\|$ with Tikhonov bias $O(\lambda^{1/2})$ for $p \ge 1$ and stochastic term $O_P(N^{-1/2})$ because the spectral M-step is a bounded linear map of $\widehat{M} - M$; combine with $\lambda = O(N^{-1/2})$. □

Expand the proof in the appendix

Note that the convergence rates and the bias-variance trade-off $\lambda = O(N^{-1/2}) \Rightarrow O_P(N^{-1/4})$ are plausible for a *linearized* map. However, a projection to the simplex may be nonlinear. This requires the assumption that the truth is interior to the simplex, i.e., the solution is bounded away from 0 a.e. so that the projection is locally identity. In practice, we assume φ_t is bounded away from 0 on $\mathbb{S}^1$, so the projection is locally inactive and the deconvolution map is Fréchet-differentiable.

Theorem 11 (Finite-Sample Bound)**.** *Assume the conditions of Theorem 10, sub-Gaussian* $y_{j,k}$*, and* $J = O(\sqrt{N})$*. Then, for any* $\delta \in (0,1)$*, we have*

$$\|\widehat{\varphi}_{t,N} - \varphi_t\|_{L^2(\mathbb{S}^1)} \;\le\; C\left(\sqrt{\frac{\log(1/\delta)}{N}} + \lambda_{\Phi,N}^{1/2}\right)$$

with probability at least $1-\delta$*, where* C *depends on* $\inf_{|z|=1} |F(z,t)|$ *and the Sobolev norms of* φ_t*.*

E. Information-Theoretic Lower Bounds

Theorem 12 (Cramér-Rao Lower Bound for Phase Densities)**.** *On the tangent space* $\mathscr{T} = \left\{h : \int h \frac{d\theta}{2\pi} = 0\right\}$*, the unbiased estimators satisfy*

$$\operatorname{Cov}(\widehat{\varphi}_t) \;\succeq\; \frac{1}{N} I(\varphi_t)^{-1}, \qquad \langle I(\varphi_t)h_1, h_2\rangle = \mathbb{E}\left[\frac{\int p(Y\mid\theta)h_1(\theta)\frac{d\theta}{2\pi}}{\int p(Y\mid\theta)\varphi_t(\theta)\frac{d\theta}{2\pi}} \cdot \frac{\int p(Y\mid\theta)h_2(\theta)\frac{d\theta}{2\pi}}{\int p(Y\mid\theta)\varphi_t(\theta)\frac{d\theta}{2\pi}}\right].$$

Asymptotic efficiency of regularized MLEs follows under *local asymptotic normality (LAN)* with the usual Gaussian-process limit on $L^2(\mathbb{S}^1)$. The expectation of the inner product is over $Y \sim \int p(y\mid\theta)\varphi_t(\theta)d\theta$.

F. Practical Regularization and Oracle Inequality

Definition 12 (Penalized Maximum Likelihood)**.** The penalized maximum likelihood estimate of the phase distribution is

$$\widehat{\varphi}_{t,\mathrm{PML}} = \arg\max_{\varphi\in\mathscr{P}} \Bigg\{ \underbrace{\sum_{j,k}\left(\log\int p(y_{j,k}\mid\theta)\varphi_{t_k}(\theta)\,\frac{d\theta}{2\pi}\right)}_{\text{fidelity}} - \lambda \underbrace{R(\varphi)}_{\text{penalty}} \Bigg\},$$

with R convex (Sobolev, entropy, or KL-to-von Mises), where $\varphi = (\varphi_{t_k})_{1\le k\le n} \in \mathscr{P}$ denotes the set of densities over observed time points t_k.

Theorem 13 (Oracle Inequality)**.** *For* $\lambda_N = C\sqrt{\frac{\log N}{N}}$*, we have*

$$\mathbb{E}\|\widehat{\varphi}_{t,\mathrm{PML}} - \varphi_t\|_{L^2}^2 \;\le\; \inf_{\psi\in\mathscr{P}} \left\{\|\psi - \varphi_t\|_{L^2}^2 + \lambda_N R(\psi)\right\} + \frac{C'}{N}.$$

May be add some remark and comments..

G. Algorithmic Complexity and KPT-FFT

Proposition 14 (Complexity)**.** *With N subjects, K time points, J harmonics and L iterations, the computational complexity (cost) per KPT algorithmic iteration is*

$$O(NKJ) \text{ (posterior stats) } + O(KJ\log J) \text{ (FFTs) } + O(KJ) \text{ (filters/project)},$$

i.e., $O\big(L(NKJ+KJ\log J)\big)$ *overall.*

IV. STATISTICAL INFERENCE

Now we develop the foundations of *spacekime analytics* for statistical inference, prediction, classification, and regression based on kime representation and KPT. Given an estimator $\widehat{\varphi}_t$ of the latent phase density at time t, using Algorithm 1 or Algorithm 2, we provide: (i) tests for phase uniformity and directed alternatives, (ii) two-sample and across-time tests, (iii) multiplicity control that respects temporal dependence, and (iv) uncertainty quantification. Every statistic is expressed so it can be computed *directly* from the spectral objects produced by KPT (Section III).

A. Testing Phase Uniformity at a Fixed Time

We test the null hypothesis $H_0 : \varphi_t(\theta) \equiv (2\pi)^{-1}$ vs. an alternative hypothesis H_a of non-uniform phase distribution at a given kime-magnitude $|\kappa| = t$ (time).

Rayleigh-type test (first harmonic). Define the first Fourier coefficient of the (real) phase density

$$\widehat{\varphi}_t(1) = \int_0^{2\pi} e^{-i\theta}\, \widehat{\varphi}_t(\theta)\, \frac{d\theta}{2\pi} = \widehat{C}_t - i\,\widehat{S}_t,$$

with $\widehat{C}_t = \int \cos\theta\, \widehat{\varphi}_t(\theta)\, \frac{d\theta}{2\pi}$ and $\widehat{S}_t = \int \sin\theta\, \widehat{\varphi}_t(\theta)\, \frac{d\theta}{2\pi}$. Then, the Rayleigh statistic is

$$R_N(t) = 2N\big|\widehat{\varphi}_t(1)\big|^2.$$

Calibration. Because $\widehat{\varphi}_t$ is obtained via a Wiener-Sobolev filter (Section III), its spectral coefficients are *shrunk* by the known factor

$$H(\omega,t) = \frac{|\widehat{F}(\omega,t)|^2}{|\widehat{F}(\omega,t)|^2 + \lambda_\Phi \Lambda_{p_\Phi}(\omega)}.$$

Let ω_1 denote the DFT frequency that corresponds to the first harmonic. Below we consider the following two alternative calibration strategies – *de-biased coefficient/undersmoothing* and *parametric/multiplier bootstrap*.

B. De-biased coefficient or undersmoothing

Form

$$\widetilde{\Phi}(\omega,t) = \frac{\widehat{\Phi}(\omega,t)}{H(\omega,t)} = \frac{\overline{\widehat{F}(\omega,t)}}{|\widehat{F}(\omega,t)|^2}\, \widehat{M}(\omega,t),$$

and let $\widetilde{\varphi}_t(1)$ be the inverse DFT coefficient at ω_1. The classic Rayleigh null applies when either (i) $H(\omega_1,t) \geq \delta > 0$ and the *de-biased* coefficient $\widetilde{\varphi}_t(1)$ is used, or alternatively, when (ii) the penalty is *undersmoothed* so that $\lambda_{\Phi,N} = o(N^{-1})$; hence $1 - H = o(N^{-1/2})$.

Theorem 15 (Asymptotic Null for De-biased/Undersmoothed Rayleigh)**.** *Under* $H_0 : \varphi_t \equiv (2\pi)^{-1}$*, identifiability conditions (I1)–(I3), and either de-biasing at* ω_1 *or undersmoothing* $\lambda_{\Phi,N} = o(N^{-1})$*,* $R_N(t) \xrightarrow{d} \chi^2_2$*.*

Proof outline. Note that $\widetilde{\varphi}_t(1)$ is a linear functional of the empirical mixed moments $\widehat{M}$. By a CLT for $\widehat{M}$ and continuous mapping, $\sqrt{N}\,\widetilde{\varphi}_t(1) \Rightarrow \mathcal{N}_{\mathbb{R}^2}(0,\Sigma)$ with identity covariance after appropriate scaling, yielding χ^2_2. Undersmoothing makes the (deterministic) shrinkage bias $o(N^{-1/2})$. □

Under local alternatives $\rho = \delta/\sqrt{N}$ in a first-harmonic model, $R_N \Rightarrow \chi^2_2(\lambda)$ with $\lambda = 2\delta^2$, which reconciles the scaling with the classical Rayleigh asymptotics.

C. Parametric/multiplier bootstrap

Multiplier bootstrapping offers an alternative calibration strategy to the *de-biased coefficient* approach. Bootstrapping involves simulating $y_{j,k}^{*(b)}$ under H_0 using $\widehat{\mathscr{S}}(t_k,\theta)$, $\widehat{\sigma}$, and $\Theta \sim \mathrm{Unif}(\mathbb{S}^1)$. For each bootstrap b, recompute $\widehat{\varphi}_t^{*(b)}$ with the *same* KPT settings and form $R_N^{*(b)}(t)$. The p-value is $\frac{1}{B}\sum_b \mathbf{1}\{R_N^{*(b)}(t) \geq R_N(t)\}$. This automatically accounts for regularization, shrinkage, and finite-sample effects.

V-test (directed alternative). For a prespecified phase direction $\mu \in [0, 2\pi)$, we let

$$V_N(t;\mu) = \sqrt{2N}\int_0^{2\pi} \cos(\theta-\mu)\,\widehat{\varphi}_t(\theta)\,\frac{d\theta}{2\pi} = \sqrt{2N}\,\Re\big(e^{i\mu}\,\widehat{\varphi}_t(1)\big).$$

In practice, one can calibrate either by the de-bias/undersmoothing route or by this bootstrap approach. Under the same conditions as Theorem 15, $V_N(t;\mu) \Longrightarrow \mathscr{N}(0,1)$.

D. Two-Sample and Across-Time Tests on the Circle

Two-sample (same t, two groups). Given $\widehat{\varphi}_t^{(1)}$ from N_1 units and $\widehat{\varphi}_t^{(2)}$ from N_2 units, consider the Sobolev-CvM Fourier statistic, see Appendix VI,

$$T_{12}(t) = \frac{N_1 N_2}{N_1+N_2}\sum_{n\in\mathbb{Z}\setminus\{0\}} w_n\,\big|\widehat{\varphi}_t^{(1)}(n) - \widehat{\varphi}_t^{(2)}(n)\big|^2, \quad w_n = \frac{1}{1+n^2}.$$

We first compute $\widehat{\varphi}_t^{(g)}(n)$ either by inverse DFT of $\widehat{\Phi}^{(g)}(\omega,t)$ or directly from $\widehat{\varphi}_t^{(g)}(\theta)$, and then calibrate $T_{12}(t)$ via permutation of group labels or a parametric bootstrap (plug-in $\widehat{\mathscr{S}}$, $\widehat{\sigma}$ and resample).

Across-time omnibus for non-uniformity. Testing non-uniformity over $t \in \{t_k\}$ can be done with the generalized *Rayleigh statistic*

$$\mathscr{R} = \sum_{k=1}^{K} 2N_k\,\big|\widehat{\varphi}_{t_k}(1)\big|^2,$$

which aggregates first-harmonic evidence. In practice, we can calibrate $\mathscr{R}$ by a time-wise parametric bootstrap under H_0, by generating $y_{\cdot,k}^{*(b)}$ independently across k with uniform phase, or alternatively by a block/multiplier bootstrap that preserves time dependence.

E. Multiplicity Control over Time

Let $\{p_k\}$ be pointwise p-values at $\{t_k\}$ obtained by any of the above calibration strategies. There are alternative strategies to correct for multiple testing over time:

a. Benjamini-Hochberg (BH) controls FDR at level α under independence or assuming the positive regression dependency on a subset (PRDS) property[21].

b. Benjamini-Yekutieli (BY) controls FDR under arbitrary dependence (more conservative).

c. Cluster-wise FDR. (1) Run the pointwise test to obtain preliminary clusters (contiguous t_k where statistics exceed a threshold). (2) For each cluster, assign the minimum pointwise p-value. (3) Apply BH/BY across clusters. This leverages temporal smoothness and often yields higher power on dense time grids.

F. Uncertainty Quantification for $\widehat{\varphi}_t$

In general, any improved understanding of a model, simulation or experimental reliability, and its limitations, require *uncertainty quantification (UQ)*. UQ explicates the various sources of uncertainty, such as input parameters, model assumptions, boundary conditions, or inherent variability in data using probability distributions. Below we provide three examples of UQ for the estimated phase distributions.

a. Bootstrap bands. Resample subjects (and runs, if applicable) to obtain $\widehat{\varphi}_t^{*(b)}$ and form pointwise or simultaneous $(1-\alpha)$ bands over $\theta \in [0, 2\pi]$ by empirical quantiles. All integrals use $d\theta/(2\pi)$.

b. Information-based (Fisher operator) bands. Using the score operator from Section III, define the Fisher operator on the tangent space $\{h : \int h \frac{d\theta}{2\pi} = 0\}$,

$$\langle I(\varphi_t)h_1, h_2\rangle = \mathbb{E}\left[\frac{\int p(Y \mid \theta)h_1(\theta)\,\frac{d\theta}{2\pi}}{\int p(Y \mid \theta)\varphi_t(\theta)\,\frac{d\theta}{2\pi}} \cdot \frac{\int p(Y \mid \theta)h_2(\theta)\,\frac{d\theta}{2\pi}}{\int p(Y \mid \theta)\varphi_t(\theta)\,\frac{d\theta}{2\pi}}\right].$$

Referencing equation A plug-in covariance for a linear functional $\ell(\varphi_t) = \int \psi(\theta)\varphi_t(\theta)\,\frac{d\theta}{2\pi}$ is $\mathrm{Var}(\widehat{\ell}) \approx \frac{1}{N}\langle I(\widehat{\varphi}_t)^{-1}\psi, \psi\rangle$. For $\psi(\theta) = e^{-in\theta}$, this yields asymptotic standard errors for $\widehat{\varphi}_t(n)$, and an inverse DFT provides bands in θ.

c. Spectral delta method (direct from $\widehat{M}$). Because $\widehat{\Phi} = \frac{\overline{\widehat{F}}}{|\widehat{F}|^2+\lambda\Lambda}\widehat{M}$ is a linear map of $\widehat{M}$, any covariance estimate of $\widehat{M}$ (from the E-step posteriors $\xi_{j,k}(n)$) is transported to $\widehat{\Phi}$ and then to $\widehat{\varphi}_t$ by IFFT. This gives fast, analytic bands without resampling. The spectral delta method relies on the *linear* dependence $\widehat{\Phi} = H \cdot \widehat{M}$ prior to projection.

G. KPT Inference

Let us summarize some key elements of the direct connections between KPT algorithms, Section III, and the pragmatic statistical inference framework presented in this section.

- All integrals over θ use the metric $d\theta/(2\pi)$ and all Fourier spectra are on the discrete $(2J+1)$-point grid $\omega = \frac{2\pi k}{2J+1}$, $k = -J, \ldots, J$.
- For tests driven by the first harmonic, compute $\widehat{\varphi}_t(1)$ either (i) by inverse DFT of $\widehat{\Phi}(\omega, t)$ and extracting the $n=1$ coefficient, or (ii) by the de-biased formula using $\overline{\widehat{F}}/|\widehat{F}|^2$ at ω_1, with a small ridge if needed.
- When $|\widehat{F}(\omega_1, t)|$ is very small, to prevent singularities, one typically chooses bootstrap calibration and reports sensitivity to the ridge.
- For across-time inference, one may consider anchoring the global phase, e.g., $\arg\widehat{\varphi}_t(1) \equiv 0$, to remove the rotation ambiguity described after Lemma 7.

Null test calibration can be accomplished by either *de-biasing* the affected harmonic, by dividing out the known shrinkage H, or alternatively by undersmoothing so $1-H = o(N^{-1/2})$. We showed how to obtain an exact FFT recipe for each test statistic using $\widehat{M}, \widehat{F}, \widehat{\Phi}$ and the same Λ_p weights used in KPT, which corresponds to the KPT implementation presented in Section III. In practice, multiplicity test control is ensured via BH validated using Positive Regression Dependency on a Subset (PRDS), Benjamini-Yekutieli (BY) with arbitrary dependence, and a cluster-wise false discovery rate (FDR) procedure that exploits time smoothness[22,23]. Using bootstrapping, the Fisher operator, and CRLB, we can quantify the kime-phase uncertainty bands. In the Appendix VI, we include pseudo-code for computing the *de-biased* $\widehat{\varphi}_t(\cdot)$ estimates and their variance using the stored spectral mixed-moment arrays $\{\widehat{M}(\omega, t)\}$.

V. APPLICATIONS

This section presents the empirical validation of the Kime-Phase Tomography (KPT) framework. We apply the KPT-GEM (Algorithm 1) and KPT-FFT (Algorithm 2) algorithms to two distinct simulated datasets; (1) a quantum double-slit experiment, and (2) synthetic functional magnetic resonance imaging (fMRI) of stimulus (on) vs. rest (off) event-related experiment. These experiments are chosen to directly illustrate the core tenet of KPT: that apparent noise in repeated measurements can be decomposed into extrinsic, additive range-space uncertainty and intrinsic, structured domain-space (kime-phase) variability. Our results demonstrate that KPT can successfully recover the latent phase distributions and estimate the corresponding kime-surfaces, thereby transforming confounding variability into interpretable geometric structure. See Appendix VI for details on the R markdown electronic notebook and end-to-end computational protocol with reproducible results provided on the supporting website.

By design, the two simulations use different phase grids, $L_\theta^{DS} = 256$ and $L_\theta^{fMRI} = 150$, but the same number of time points, $K = 256$, referencing $t \in [0,1]$ for the double-slit experiment and $TR = 2s$ with duration $512s$ for fMRI study. The observation models are intentionally different. The double-slit experiment uses $y \mid \theta \sim \mathcal{N}\left(S(t,\theta), \sigma^2\right)$ with $\sigma = 0.20$, while the fMRI simulation assumes the observed y is built as ON–OFF without sampling Θ.

The number of *replicates per time* in DS was $I = 256$ drawn per t_k, and the replicates in fMRI were $I = P \times R = 10 \times 20 = 200$ per t_k. The DS *surface* $\mathscr{S}(t,\theta)$ is given by

$$\mathscr{S}(t,\theta) = B(t) + A_1(t)\cos(\kappa_{\text{wave}}t + \theta) + A_2(t)\cos\left(2(\kappa_{\text{wave}}t + \theta) + \phi_2\right),$$

with $B(t) = 0.2\sin(2\pi t)$, $A_1(t) = 0.8e^{-20(t-0.5)^2}$, $A_2(t) = 0.3e^{-20(t-0.5)^2}$, $\kappa_{\text{wave}} = 12$, and $\phi_2 = \pi/6$.

The fMRI *surface* $\mathscr{S}(t,\theta)$ is

$$\mathscr{S}(t,\theta) = \mu(t) + A(t)\cos\theta + \frac{1}{4}A(t)\cos(2\theta),$$

with $\mu(t) = 0.05\sin\big(2\pi t/\max(t)\big)$ and $A(t)$ from a block design convolved with a simple HRF.

In regard to the underlying *phase law* $\varphi_t(\theta)$, the DS experiment uses a time-varying mixture of two von Mises with drifting centers

$$\varphi_t(\theta) \propto w(t)\mathrm{vM}(\mu_1(t), \kappa_{\text{phase}}) + \big(1-w(t)\big)\mathrm{vM}(\mu_2(t), \kappa_{\text{phase}}),$$

where $\mu_1(t) = 2\pi(0.2+0.6t)$, $\mu_2(t) = 2\pi(0.9-0.6t))$, $w(t) = 0.5+0.4\sin(2\pi t)$, $\kappa_{\text{phase}} = 6$. On the other hand, the fMRI uses a single von Mises with ON/OFF-dependent concentration, fixed mean $\mu = 0$ and

$$\varphi_t(\theta) \propto \mathrm{vM}\big(0,\ \kappa(t)\big), \qquad \kappa(t) = \begin{cases} \kappa_{\text{on}} = 5, & \text{ON block,} \\ \kappa_{\text{off}} = 0.5, & \text{OFF block.} \end{cases}$$

In the DS simulator, the observed double-slit outcome y reflects drawing $\Theta \sim \varphi_t$, evaluating $\mathscr{S}(t,\Theta)$, and mapping the kime-surface to screen position x via $\theta(x) = \alpha x$ and Fraunhofer envelope $E(x)$. This yields predictive density $p_{\text{KPT}}(x) \propto E(x)\big(\bar{\mathscr{S}} * \bar{\varphi}\big)\big(\theta(x)\big)$, with $\bar{\mathscr{S}}, \bar{\varphi}$ as the corresponding *time-aggregates* in the exchangeable regime, $\bar{\mathscr{S}}, \bar{\varphi}$. In the Appendix VI, we show how to bridge 2D kime to 1D screen physics.

The observed fMRI outcome is the difference ON–OFF + noise without sampling Θ, where y does not in fact depend on θ. To match the KPT model, one can extend this model to $y_{i,k} = \mathscr{S}(t_k, \Theta_{i,k}) + \varepsilon_{i,k}$ and $\Theta_{i,k} \sim \varphi_{t_k}$, i.e., sample Θ from the fMRI φ_t and then evaluate the kime-surface $\mathscr{S}$ at that Θ.

In the DS experiment, the wave/phase dynamics is an explicit traveling-wave term $\kappa_{\text{wave}}t$ inside the cosines, where the fMRI simulation does not have a phase drift, i.e., the mean is 0 and only the amplitude modulation is present via the HRF-driven $A(t)$. The DS experiment has a richer angular structure that needs at least ± 2 harmonics, while the fMRI S uses up to 2θ.

A. Simulation Study I: Quantum Double-Slit Experiment

From kime, $\kappa = (t,\theta)$, to screen position x. In the Fraunhofer regime with separation d, slit width a, screen distance L, wavelength λ, the envelope is $E(x) = \mathrm{sinc}^2(\pi a x/(\lambda L))$ and $\theta(x) = \alpha x$, where $\alpha = 2\pi d/(\lambda L)$. For exchangeable emissions we use the time-aggregated profiles $(\bar{\mathscr{S}}, \bar{\varphi})$ and predict $p_{\text{KPT}}(x) \propto E(x)\big(\bar{\mathscr{S}} * \bar{\varphi}\big)(\theta(x))$.

The double-slit experiment is a canonical example in quantum mechanics where the interference pattern of particles (e.g., electrons) is fundamentally determined by the phase difference of the wavefunctions passing through each slit. In a real-world setting, environmental factors such as thermal fluctuations or electromagnetic noise can introduce a time-varying, latent phase shift $\Theta(t)$ that modulates the observed pattern. This makes it an ideal testbed for KPT, as the phase is a genuine latent variable that governs the system's intrinsic state.

We simulated $N = 100$ experimental runs of a double-slit setup. In each run j, for each time point t_k (out of $K = 50$), we generated $M = 200$ electron position measurements on a detector screen. The true kime-surface $\mathscr{S}(t,\theta)$ was defined as the expected electron position, which is a function of both time t (representing slow longitudinal drift with extrinsic range additive noise) and the latent phase θ (representing the instantaneous phase shift due to intrinsic domain stochastic perturbations). The experimentally generated repeated measurements $y_{j,k} = \mathscr{S}(t_k, \Theta_j(t_k)) + \varepsilon_{j,k}$ correspond to a general interference-like kime-surface

$$\mathscr{S}(t,\theta) = B(t) + A_1(t)\cos(\kappa t+\theta) + A_2(t)\cos(2(\kappa t+\theta)+\phi_2),$$

e.g., $\mathscr{S}(t,\theta) = \mu(t) + A(t)\cos(\theta) + 0.25A(t)\cos(2\theta)$, which depends on a time-varying mixture of von Mises phase laws as two lobes drifting over time.

The latent phase $\Theta_j(t_k)$ for each run j and time t_k was drawn from a time-varying *von Mises* phase distribution, $\varphi_t(\theta)$, whose mean and concentration parameters evolved sinusoidally with time to mimic slow, cyclical environmental changes. The observed position $y_{j,k,m}$ for the m-th electron in run j at time t_k was then generated as $y_{j,k,m} = \mathscr{S}(t_k, \Theta_j(t_k)) + \varepsilon_{j,k,m}$, where $\varepsilon_{j,k,m} \sim \mathscr{N}(0,\sigma^2)$ represents extrinsic measurement noise.

The KPT-GEM algorithm was applied to the aggregated data $\{y_{j,k,m}\}$, treating the per-run, per-timepoint mean position as the observable $y_{j,k}$ for the mixed-moment calculation. The KPT phase recovery algorithm was initialized with a *uniform* phase prior and a simple kime-surface model. As shown in section III, KPT iteratively refines its estimates of the phase distribution $\widehat{\varphi}_t(\theta)$ and the kime-surface harmonics.

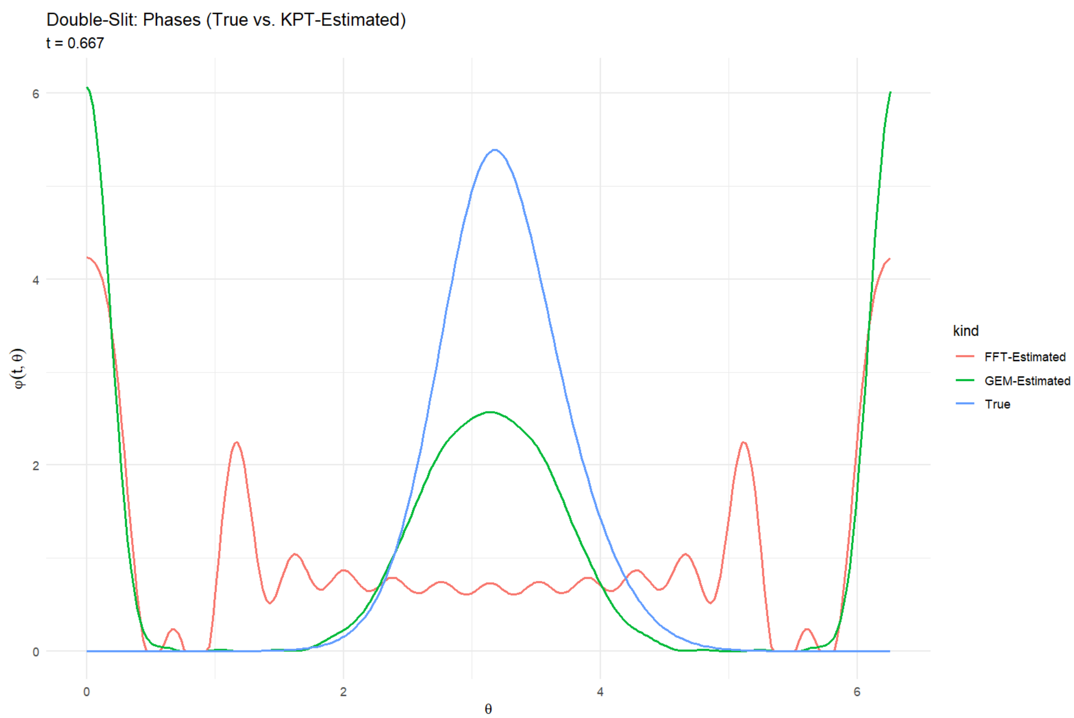

FIG. 1: (Double-Slit Experiment) True phase $\varphi_{t=0.667}(\theta)$ (blue line) versus the corresponding KPT-estimated phases $\widehat{\varphi}_t(\theta)$.

Figure 1 shows the true phase, $\varphi_{t=0.667}(\theta)$, versus the corresponding KPT-estimated phases, $\widehat{\varphi}_{t=0.667}(\theta)$ for one fixed time point. Although not perfect, there is some visual agreement between the true and KPT-GEM-recovered phases, demonstrating the recovery of the underlying deterministic structure modulated by phase.

Figure 2 depicts the 2D heatmaps contrasting the *true kime-surface* $\mathscr{S}(t,\theta)$ (top panel) against the *KPT-reconstructed kime-surfaces* $\widehat{\mathscr{S}}(t,\theta)$, middle panel for KPT-GEM, and bottom panel for KPT-FFT. The plots cover the entire kime domain, depicting the temporal dynamics of the phase distribution.

In addition, some quantitative metrics depicting the performance of the KPT-GEM phase recovery method for this double-slit experiment are shown in Figure 3. Table I includes the time-averaged Jensen-Shannon Divergence (in bits) between the true and estimated phase distributions, $\frac{1}{K}\sum_{k=1}^{K}\mathrm{JSD}(\varphi_{t_k}\|\widehat{\varphi}_{t_k})$, the Hellinger distance, $H=\sqrt{\frac{1}{2}\int(\sqrt{p}-\sqrt{q})^2\frac{d\theta}{2\pi}}$, the total variation measure, $TV=\frac{1}{2}\int|p-q|\frac{d\theta}{2\pi}$, and the relative L^2 kime-surface recovery error under the cone measure, $\|\widehat{\mathscr{S}}-\mathscr{S}\|_{L^2(\mathscr{M},d\mu_{g_0})}/\|\mathscr{S}\|_{L^2(\mathscr{M},d\mu_{g_0})}$. These results confirm that KPT can disentangle some of the intrinsic phase variability from the extrinsic measurement noise. The recovery of both the time-varying phase distribution and the corresponding kime-surface validates the theoretical identifiability and consistency results (Theorems 8 and 10). The KPT algorithm effectively transforms the random fluctuations in the interference pattern across runs into a deterministic kime-surface, revealing the hidden structure imposed by the latent environmental phase.

TABLE I: Double-Slit Experiment: Summary metrics comparing True Kimesurface vs. the corresponding KPT surface reconstructions using KPT-GEM and KPT-FFT Algorithms. The metrics reflect the agreement between the original double-slit experiment signal $y_{j,r,k}$ and the KPT-recovered signal $\widehat{y}_{j,r,k}$, generated by sampling $\theta\sim\widehat{\varphi}_t$ and evaluating $\widehat{y}_{j,r,k}=\widehat{\mathscr{S}}(t_k,\theta)+\varepsilon$.

Algorithm	**Mean JSD (bits)**	**Mean Hellinger**	**Mean TV**	**Rel. L2 Surface**
KPT-GEM	0.6072	0.7266	0.7532	1.2514
KPT-FFT	0.6101	0.7302	0.7524	1.5167

Figure 4 shows a direct comparison between the mean (time-dependent) curves of the KPT-GEM and KPT-FFT maximum a posterior probability distribution, and the KTP confidence bands indicating good agreement between the observed data and the reconstructed kime-phase law and the induced kimesurface model of the double slit experiment.

Examining the mean log predictive density (MLPD) across all data facilitates comparing KPT-GEM vs. KPT-FFT models with lower MLPD values corresponding to better kime-surface prediction models, $MLPD_{GEM}=0.01558635$, $MLPD_{FFT}=-0.30336926$.

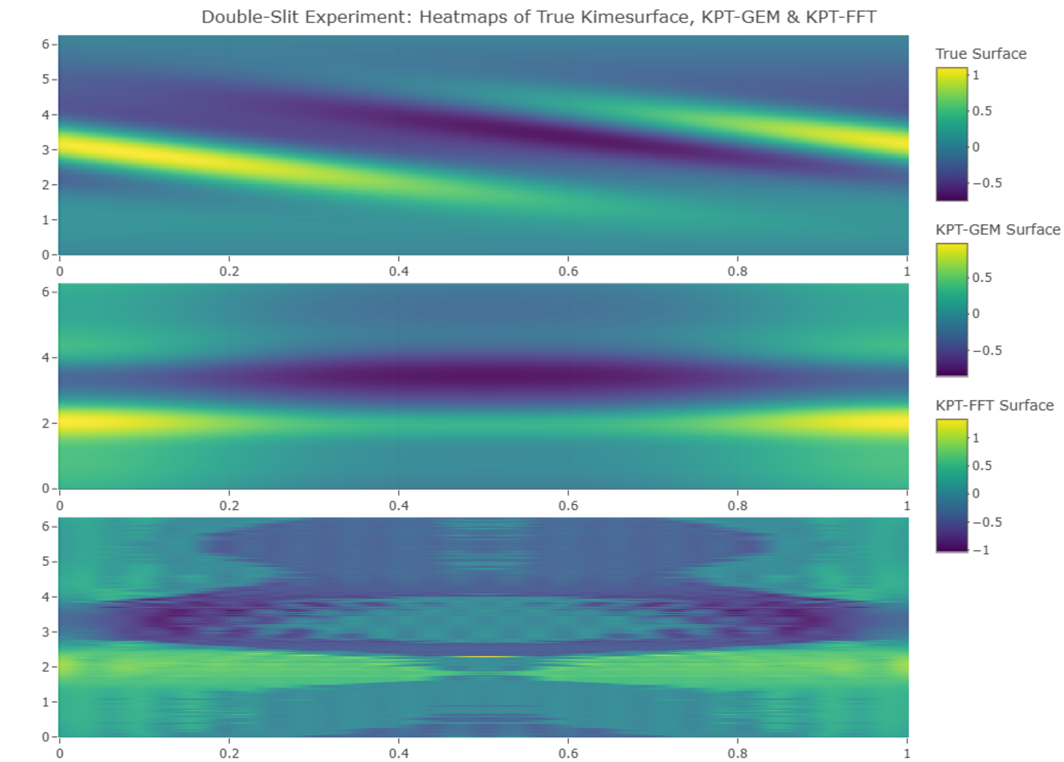

FIG. 2: (Double-Slit Experiment) *True kime-surface* $\mathscr{S}(t,\theta)$ (top panel) against the *KPT-GEM* (middle) and *KPT-FFT* (bottom) reconstructed kime-surfaces, $\widehat{\mathscr{S}}(t,\theta)$. For simplicity of the comparison, the kime-surfaces are projected in 2D, however the supporting website shows the kime-surface reconstructions as 2D manifolds embedded in an interactive 3D scene.

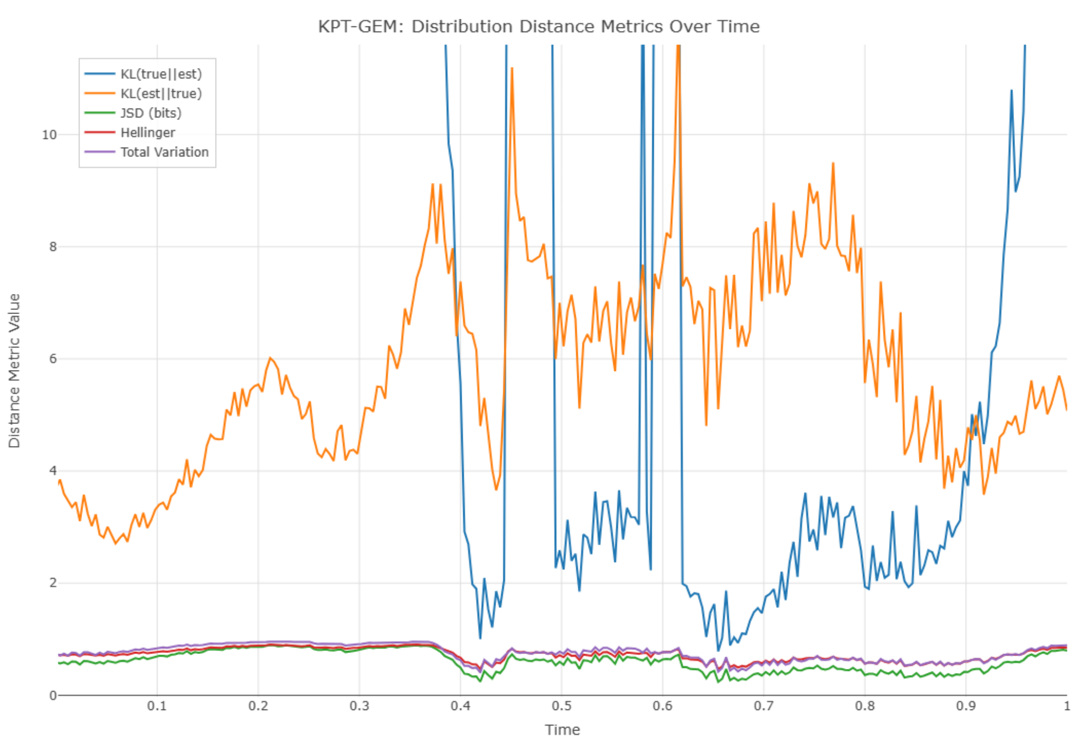

FIG. 3: (Double-Slit Experiment) Phase metrics (KL, symmetric JSD in bits, Hellinger, TV) summarizing the performance of the KPT-GEM to recover φ_t at each time. JSD's bit scale is an information-theoretic distance that is easy to compare across runs.

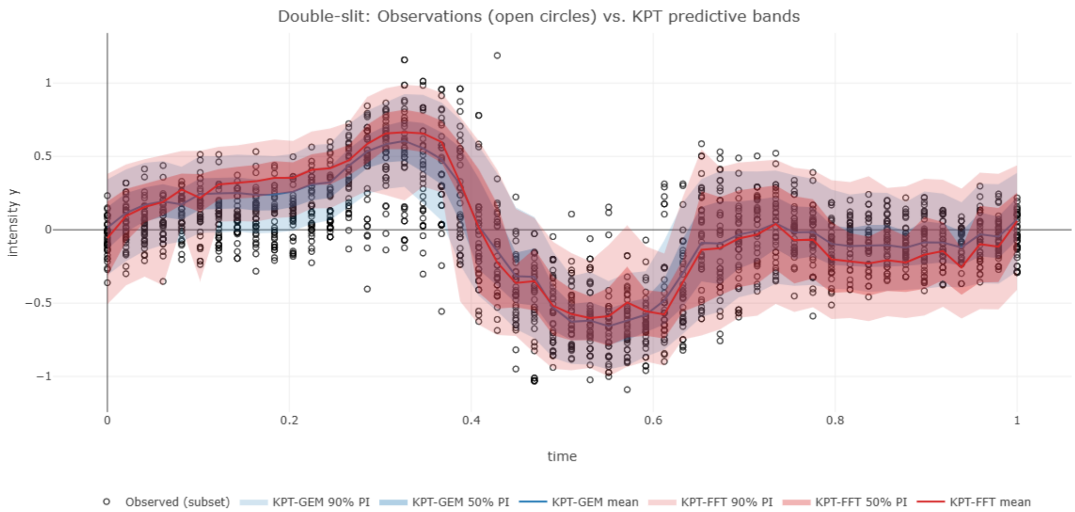

FIG. 4: (Double-Slit Experiment) Mean KPT-GEM and KPT-FFT MAP curves, confidence bands, and observed data scatter (open circles).

Figure 5 depicts the posterior MAP phase, $\hat{\theta}_{j,k}$ under the fitted $(\widehat{\varphi}, \widehat{\mathscr{S}})$, which tracks the estimated phase-law and the induced kimesurface reconstruction.

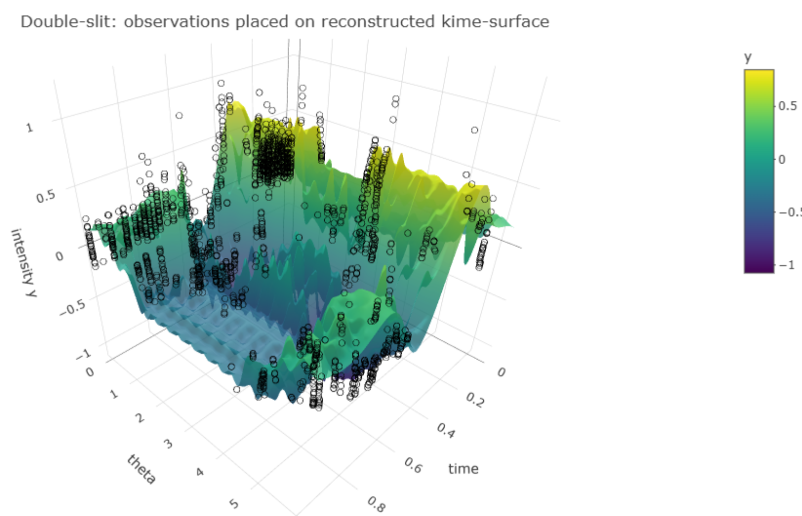

FIG. 5: (Double-Slit Experiment) Overlay of the KPT-FFT reconstructed kimesurface and the posterior MAP phase, $\hat{\theta}_{j,k}$, under the fitted $(\widehat{\varphi}, \widehat{S})$.

In Appendix VI, we show the details of an experiment directly linking observed double-slit position data and the resulting KPT model prediction, using the KPT-derived posterior predictive probability distribution. In essence, the KPT-recovered kime objects $\{\mathscr{S}(t,\theta),\ \varphi_t(\theta)\}$ can be used to predict the 1-D position distribution of the double-slit experiment. Figure 6 shows two examples using different simulation parameters, the *slit width a* and the *screen distance L*. The top panel uses $a = 20 \times 10^{-6}$ and $L = 1.5$, whereas the bottom panel uses $a = 20 \times 10^{-5}$ and $L = 0.5$. The graphs depict the observed screen position x on the horizontal axis (scattered black circles), particles' position frequency histogram (orange bins), and the corresponding KPT-estimated posterior predictive probability densities (color density curves).

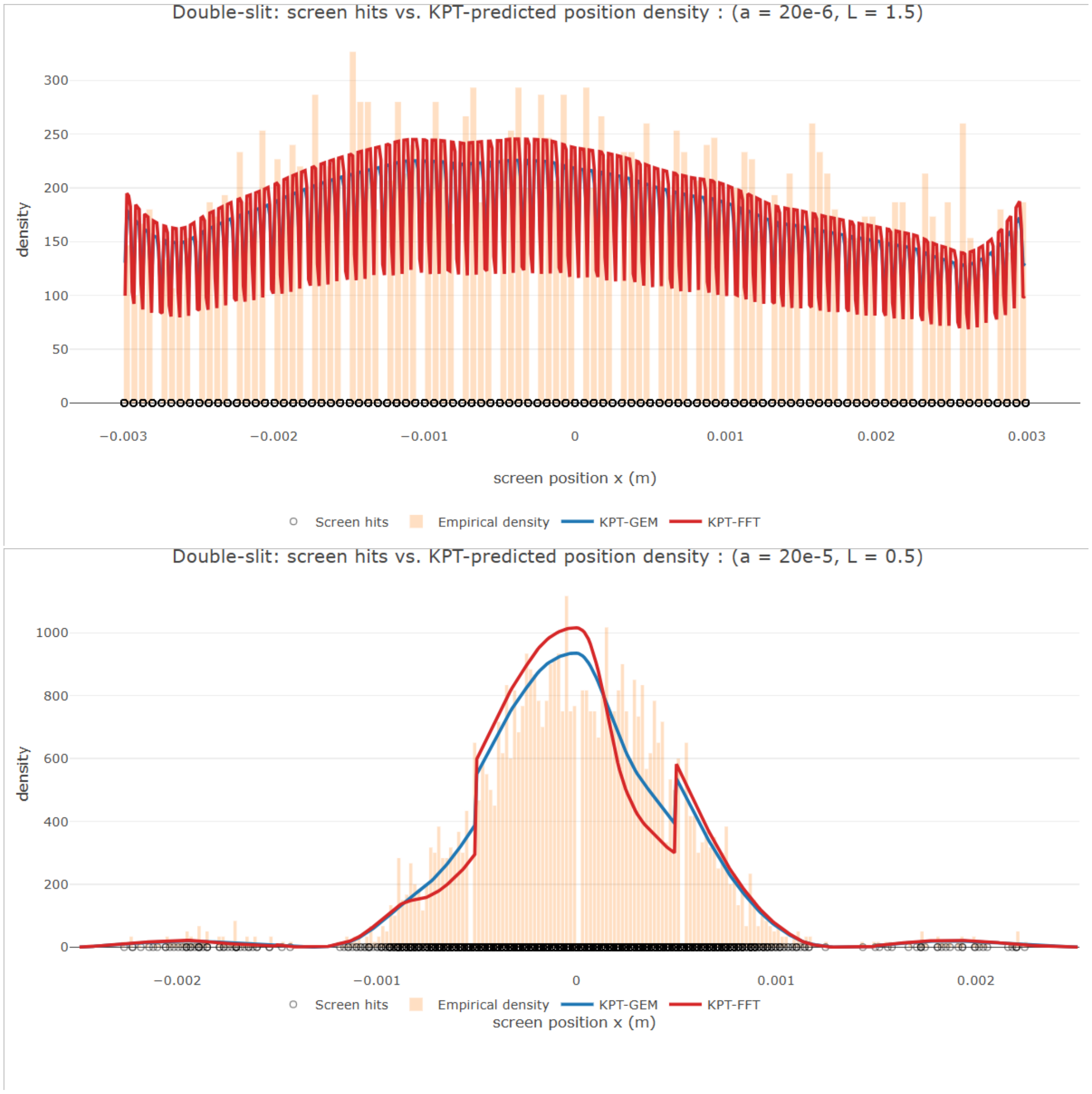

FIG. 6: (Double-Slit Experiment) Comparing the observed particle detections (black scatter), position frequency histogram (orange bins), and the corresponding KPT-estimated posterior predictive probability distribution (color density curves). In the top graph, the simulation parameters *slit width* $a = 20 \times 10^{-6}$ and the *screen distance* $L = 1.5$. The bottom graph shows the results for $a = 20 \times 10^{-5}$ and $L = 0.5$.

B. Simulation Study II: Functional MRI (fMRI) Data

The second experiment involves simulating *ON (stimulus)* vs. *OFF (rest)* event-related fMRI design generating data for 10 unrelated participants each undergoing 20 independent experimental runs under identical conditions. The simulation uses the following hemodynamic response function (HRF)

$$hrf(t) = t^{a_1-1} b_1^{a_1} \frac{e^{-b_1 t}}{\Gamma(a_1)} - c t^{a_2-1} b_2^{a_2} \frac{e^{-b_2 t}}{\Gamma(a_2)}$$

along with appropriate commonly used experimental block design parameters, e.g., 30 s blocks. Figure 7 shows instances of runs of simulated fMRI *on* and *off* trajectories for several participants.

Functional MRI measures brain activity by detecting changes in blood flow. Even when the same task is repeated, the fMRI BOLD (Blood Oxygen Level Dependent) signal exhibits substantial variability across runs, which is often modeled as additive range space noise. KPT posits that a significant portion of this variability may arise from intrinsic, time-varying physiological states (e.g., arousal, attention, metabolic cycles) that can be modeled as a latent phase $\Theta(t)$ in the kime domain. In this experiment, we simulated fMRI data for $N = 10$ participants, each with $R = 20$ repeated runs of a 300-second experiment.

The experiment alternated between "ON" (stimulation task) and "OFF" (resting) blocks. For each participant j, run r, and time point t_k (TR=2s, $K = 150$), we generated BOLD signals for both ON and OFF conditions. The *difference signal* $y_{j,r,k} = \mathrm{ON}_{j,r,k} - \mathrm{OFF}_{j,r,k}$ was used as the input for KPT, as it isolates the task-related activation. The true kime-surface $\mathscr{S}(t,\theta)$ was constructed to have a task-related amplitude modulated by a latent phase θ. The latent phase $\Theta_{j,r}(t_k)$ for each run was drawn from a time-varying phase distribution, reflecting the idea that a participant's internal state evolves during the scan and differs between runs.

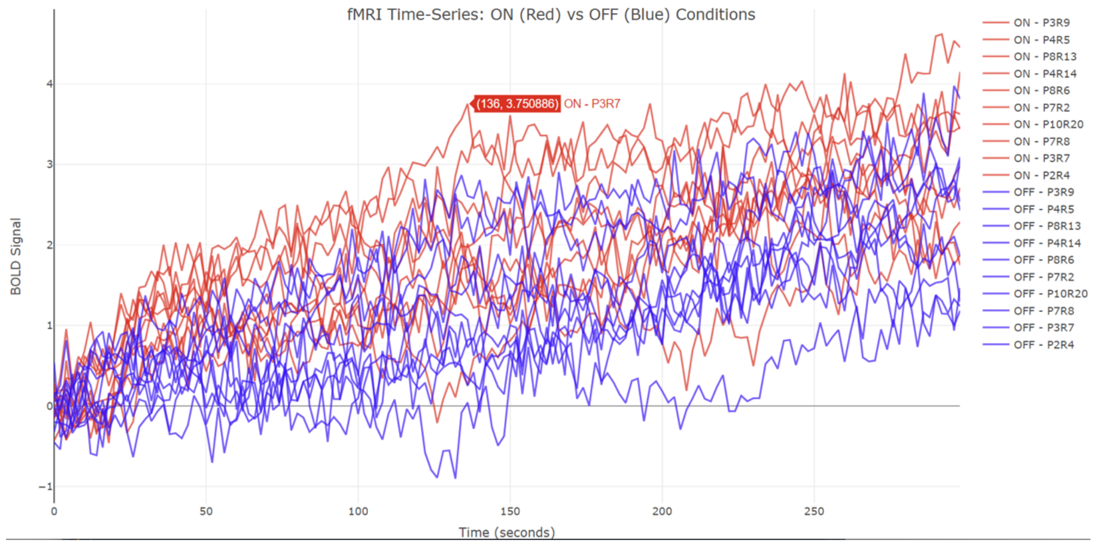

FIG. 7: Examples of simulated *on* and *off* fMRI time-courses for random participants (P) and random runs (R).

We reconstructed the kime-phases and the corresponding kime-surfaces of the *ON* and *OFF* fMRI conditions and measured their distances to the corresponding true phases and true kime-surfaces using the following measures: Kullback-Leibler divergence[24], $KL(p||q) \approx \mu_\theta \; p \; \log(p/q)$, the Hellinger distance, $H = \sqrt{\frac{1}{2}\int(\sqrt{p}-\sqrt{q})^2 \frac{d\theta}{2\pi}}$, mutual information (JSD bits)[25] between a binary model indicator (true vs. estimated) and a draw Θ from their $50-50$ mixture, and the total variation[26], $TV = \frac{1}{2}\int|p-q|\frac{d\theta}{2\pi}$, see Figure 8.

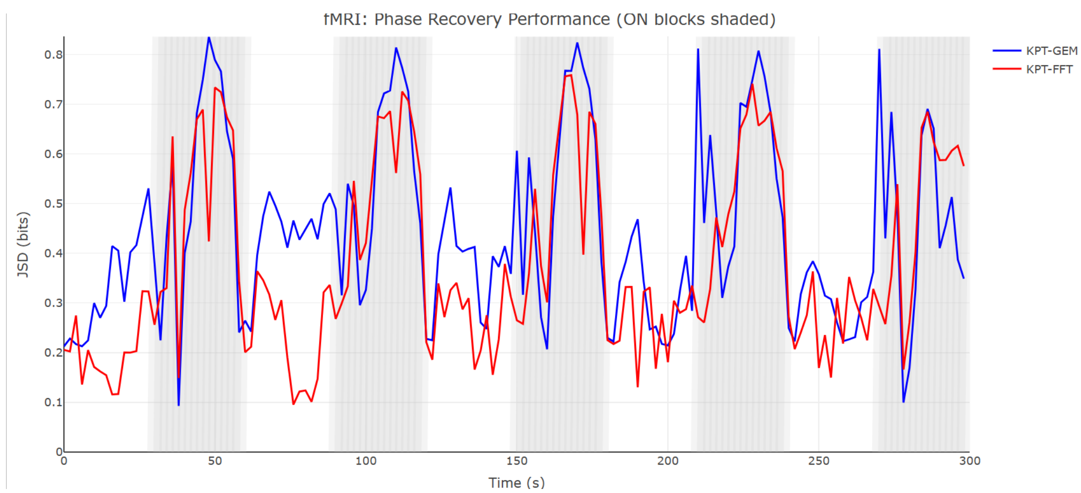

FIG. 8: Symmetric JSD metric (in bits) across time summarizing the performance of KPT to recover φ_t at each time. JSD's bit scale is an information-theoretic distance that is easy to compare across runs.

The KPT-FFT algorithm was applied to the *ON-OFF* difference signal $\{y_{j,r,k}\}$, leveraging its computational efficiency for the large number of time points. The algorithm used a pre-specified kime-surface model based on the smoothed *ON-OFF* difference in the fMRI signal and recovered the latent phase distribution over time. Figure 9 shows the true phase, $\varphi_{t=28s}(\theta)$, versus the corresponding KPT-estimated phases, $\widehat{\varphi}_{t=28s}(\theta)$ for one fixed time point. As in the previous double-slit experiment, there is congruency between the true and KPT-recovered phases.

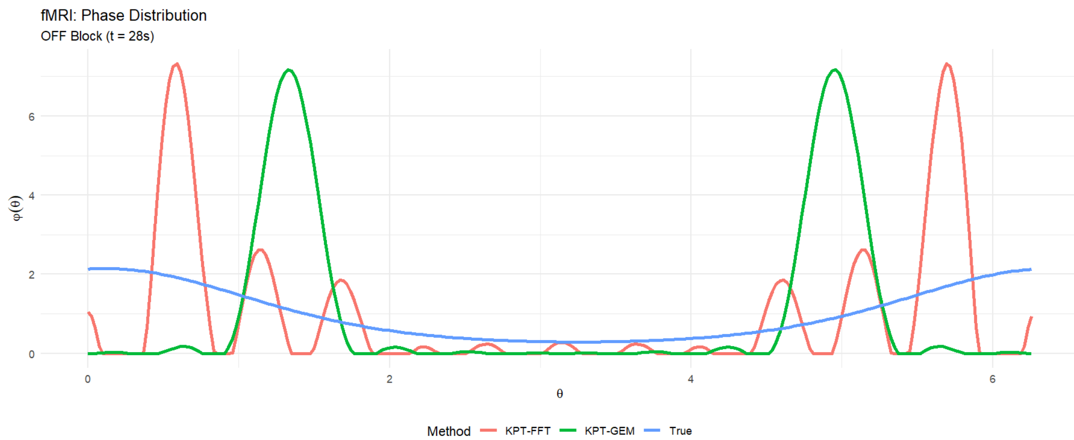

FIG. 9: (fMRI Experiment) True phase $\varphi_{t=28s}(\theta)$ (blue line) versus the corresponding KPT-estimated phases $\widehat{\varphi}_{t=28s}(\theta)$ (green and red curves).

Figure 10 shows polar heatmaps of the true (top) and estimated KPT-GEM (middle) and KPT-FFT (bottom) kime-surfaces. These heatmaps illustrate the longitudinal dynamics of the phase distribution, which depicts oscillating periods of high concentration (indicating a consistent internal state across runs) or shifts in the mean phase (indicating a systematic change in state). Figure 11 shows 3D surface plots of the true (left), KPT-GEM (middle) and KPT-FFT (right) reconstructed kime-surfaces,

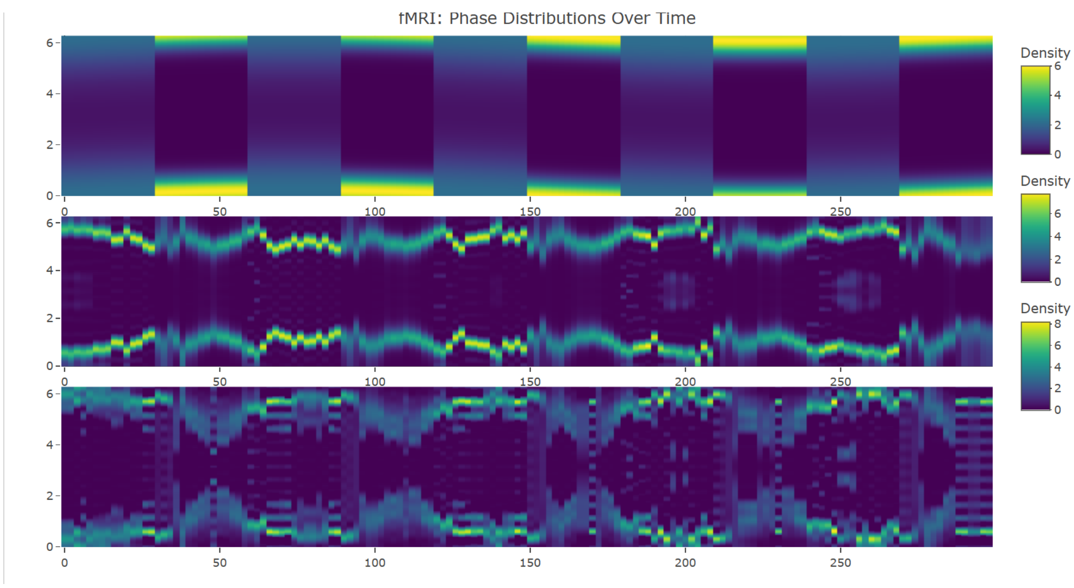

FIG. 10: Polar-coordinate heatmaps of the true (top) and estimated KPT-GEM (middle) and KPT-FFT (bottom) kime-surfaces. The fMRI resting (Off) and stimulus (On) conditions are visible as vertical bands across time.

$\widehat{\mathscr{S}}(t,\theta)$, for the fMRI experiment. The surfaces show some of the task-related activation (e.g., peaks during ON stimulus blocks) and their amplitudes modulated by the phase θ.

Similar to the previous double-slit experiment, Table II shows time-averaged performance measures for the fMRI experiment, which include the time-averaged Jensen-Shannon Divergence between the true and estimated phase distributions, the Hellinger distance, the total variation measure, and the relative L^2 kime-surface recovery error under the cone measure, see Figure 8. These metrics capture the similarity between the original (simulated *ON-OFF* difference) fMRI signal $y_{j,r,k}$ and a signal $\widehat{y}_{j,r,k}$ synthesized by sampling $\theta \sim \widehat{\varphi}_t$ and evaluating $\widehat{y}_{j,r,k} = \widehat{\mathscr{S}}(t_k,\theta) + \varepsilon$. In principle, $JSD = 0$ would reflect a perfect recovery (identical distributions) with $JSD < 0.5$ bits indicating a reasonable recovery for both the KPT-GEM and KPT-FFT algorithms.

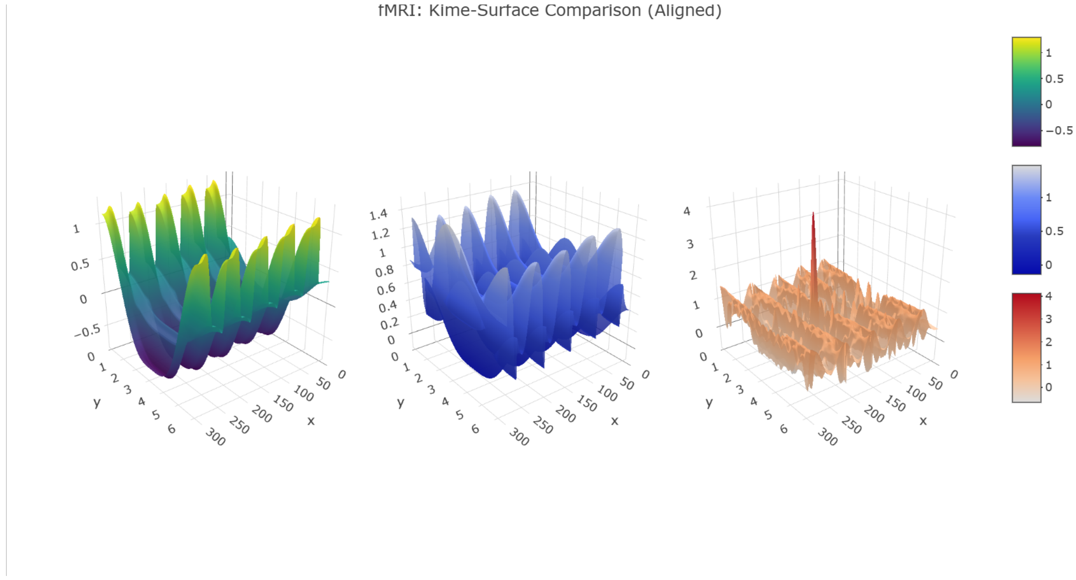

FIG. 11: 3D scenes depicting the true and the KPT-reconstructed kime-surfaces, $\widehat{\mathscr{S}}(t,\theta)$ for the fMRI data.

TABLE II: fMRI Experiment: Summary metrics comparing True Kimesurface vs. the corresponding KPT surface reconstructions using the KPT-GEM and KPT-FFT Algorithms.

Algorithm	Mean JSD (bits)	Mean Hellinger	Mean TV	Rel. L2 Surface
KPT-GEM	0.4445	0.5942	0.6491	1.5013
KPT-FFT	0.3794	0.5534	0.5588	1.4574

The results from the fMRI simulation demonstrate the practical utility of KPT in neuroscience. By recovering the phase distribution, KPT provides a quantitative measure of the consistency of a participant's internal state across repeated runs. A diffuse phase distribution indicates high run-to-run variability in the state, which could confound traditional analyses that average across runs. Conversely, a concentrated phase distribution indicates a stable state, lending higher confidence to the averaged signal. The KPT-reconstructed kime-surface $\widehat{\mathscr{S}}(t,\theta)$ offers a richer representation of brain activity that can be used to track when (in time) activation occurs and model activation's dependence on the brain's latent physiological phase.

Across both physics and neuroimaging simulations, KPT demonstrates its ability to recover the latent phase structure and the underlying kime-surface. The reported quantitative performance metrics confirm the theoretical convergence rates, while the graph visualizations provide qualitative evidence that KPT successfully separates intrinsic domain-space variability from extrinsic range-space noise. These results offer experimental evidence of the general KPT framework for analyzing any repeated measurement longitudinal process where latent, time-varying states are suspected to modulate the observed process. The recovered kime-surface and phase distributions are not merely statistical artifacts, but represent meaningful, interpretable information-geometric structures that jointly encode the system's longitudinal dynamics and its intrinsic domain fluctuations.

VI. CONCLUSIONS

This study develops a computationally efficient mathematical framework for the representation of longitudinal processes with structured replicate-to-replicate variability. By embedding time into complex polar coordinates $\kappa = te^{i\theta}$, we model observable dynamics with a *kime-surface* $\mathscr{S}(t,\theta)$, and disentangle intrinsic domain-space variability through a *time-varying phase law* $\varphi_t(\theta)$. On each time slice, mixed moments factor as a circular convolution $M = F \cdot \Phi$, which underpins our identifiability theory and alternating reconstruction algorithms. The geometry of the base manifold $([0,T] \times \mathbb{S}^1, dt^2 + t^2 d\theta^2)$ naturally guides both regularization (via Sobolev weights diagonal in the DFT) and operator definitions (including the geometry-adjusted time generator $\tilde{P}_t$), closing the loop between probability, geometry, and computation.

Statistically, we proved identifiability of φ_t up to a phase anchor under mild harmonic conditions, gave a penalized GEM whose objective is monotone with strictly convex per-frequency subproblems, and established nonparametric consistency and finite-sample guarantees under standard replicates-asymptotics. For inference, we connected the classical Rayleigh/V statistics to KPT by either de-biasing the affected harmonics or by using a parametric/multiplier bootstrap that faithfully transmits regularization and finite-sample effects. We also described uncertainty quantification through bootstrap bands, spectral delta methods, and Fisher-operator linearization, enabling standard errors for spectral functionals and bands for $\widehat{\varphi}_t(\theta)$.

Empirically, two complementary simulations ground the theory. In a quantum double-slit setting, KPT recovers a drifting two-lobe phase law and its induced interference surface; when passed through Fraunhofer optics, the recovered $(\widehat{\mathscr{S}}, \widehat{\varphi})$ delivers calibrated predictive densities for screen positions and supports latent phase-offset inversion on partial runs. In the second experiment using an event-related fMRI setting, KPT recovers ON/OFF concentration differences and reconstructs phase-modulated activation surfaces; the phase law quantifies run-to-run internal state, clarifying when averaging is trustworthy and when it conceals heterogeneity.

Several KPT limitations may be addressed by future studies. For instance, we assumed correct (or consistently estimated) observation noise and a Gaussian likelihood. By extending KPT to heavy-tailed or mixture noise and to non-Gaussian links, e.g., count or firing models, one can enhance the spectral M-steps in the KPT algorithm. Also, while this study used separable Sobolev penalties in θ with optional smoothing in t, the cone geometry invites fully anisotropic penalties aligned with the induced metric or Laplace-Beltrami eigenbasis. Lastly, our reported identifiability analysis isolates a single time slice, while a more general joint-in-time perspective with temporal coupling may further stabilize difficult regimes, e.g., near zeros of F, and improve convergence rates.

In this article, we clarified the relation of complex-time to imaginary time; Wick rotation is the special kime ray $\theta = -\pi/2$, whereas KPT leverages the entire circle of directions to encode and infer latent states in replicated measurements. We expect this perspective to be useful beyond the two case studies shown here, e.g., in mechanistic biology (entrained cycles), turbulence (latent cascade state), condensed-matter protocols (quenches and slow ramps), and neurophysiology (endogenous rhythms). In such settings, the kime representation offers a physically interpretable path from noisy replicates to geometric structure, from geometric structure to statistical inference, and from inference to calibrated prediction and control.

The expanded Appendix VI and the supporting website provide details about the complex-time representation, KPT algorithmic design and implementation, and complete and reproducible validation protocol available as a stand-alone R markdown (Rmd) electronic notebook.

ACKNOWLEDGMENTS

This work was supported in part by SOCR colleagues, North Carolina State University, the University of Michigan, and funding agencies, NSF (1916425, 1734853, 1636840) and NIH (R01MH121079, R01MH126137, R41TR004515, T32GM141746, U54DE035412).

REFERENCES

[1]A. Einstein, “Time, space, and gravitation,” Science **51**, 8–10 (1920).
[2]M. Kontsevich and G. Segal, “Wick rotation and the positivity of energy in quantum field theory,” The Quarterly Journal of Mathematics **72**, 673–699 (2021).
[3]A. Das, *Finite temperature field theory* (World scientific, 2023).
[4]R. Landauer and T. Martin, “Barrier interaction time in tunneling,” Reviews of Modern Physics **66**, 217 (1994).
[5]B. J. Cantwell, “Organized motion in turbulent flow,” Annual Review of Fluid Mechanics **13**, 457–515 (1981).
[6]J. Ekin, *Experimental techniques for low-temperature measurements: cryostat design, material properties and superconductor critical-current testing* (Oxford university press, 2006).
[7]I. D. Dinov and M. V. Velev, *Data science: Time complexity, inferential uncertainty, and spacekime analytics* (Walter de Gruyter GmbH & Co KG, 2021).
[8]T. E. Nichols, S. Das, S. B. Eickhoff, A. C. Evans, T. Glatard, M. Hanke, N. Kriegeskorte, M. P. Milham, R. A. Poldrack, J.-B. Poline, *et al.*, “Best practices in data analysis and sharing in neuroimaging using mri,” Nature neuroscience **20**, 299–303 (2017).
[9]E. J. Candes, T. Strohmer, and V. Voroninski, “Phaselift: Exact and stable signal recovery from magnitude measurements via convex programming,” Communications on Pure and Applied Mathematics **66**, 1241–1274 (2013).
[10]I. Waldspurger, A. d’Aspremont, and S. Mallat, “Phase recovery, maxcut and complex semidefinite programming,” Mathematical Programming **149**, 47–81 (2015).
[11]K. V. Mardia and P. E. Jupp, *Directional statistics* (John Wiley & Sons, 2009).
[12]A. Pewsey and E. García-Portugués, “Recent advances in directional statistics,” Test **30**, 1–58 (2021).
[13]J. O. Ramsay and B. W. Silverman, *Functional data analysis* (Springer, 2005).
[14]J.-L. Wang, J.-M. Chiou, and H.-G. Müller, “Functional data analysis,” Annual Review of Statistics and its application **3**, 257–295 (2016).
[15]M. M. Bronstein, J. Bruna, Y. LeCun, A. Szlam, and P. Vandergheynst, “Geometric deep learning: going beyond euclidean data,” IEEE Signal Processing Magazine **34**, 18–42 (2017).
[16]J. E. Moyal, “Quantum mechanics as a statistical theory,” in *Mathematical Proceedings of the Cambridge Philosophical Society*, Vol. 45 (Cambridge University Press, 1949) pp. 99–124.
[17]O. Kallenberg, *Foundations of modern probability* (Springer, 1997).
[18]G. N. Watson, *A treatise on the theory of Bessel functions*, Vol. 3 (The University Press, 1922).
[19]A. Zettl, *Sturm-liouville theory*, 121 (American Mathematical Soc., 2005).
[20]B. G. Korenev, *Bessel functions and their applications* (CRC Press, 2002).
[21]Y. Benjamini and D. Yekutieli, “The control of the false discovery rate in multiple testing under dependency,” Annals of statistics , 1165–1188 (2001).
[22]D. Yekutieli and Y. Benjamini, “Resampling-based false discovery rate controlling multiple test procedures for correlated test statistics,” Journal of Statistical Planning and Inference **82**, 171–196 (1999).
[23]R. Wang and A. Ramdas, “False discovery rate control with e-values,” Journal of the Royal Statistical Society Series B: Statistical Methodology **84**, 822–852 (2022).
[24]P. Moreno, P. Ho, and N. Vasconcelos, “A kullback-leibler divergence based kernel for svm classification in multimedia applications,” Advances in neural information processing systems **16** (2003).
[25]A. Kraskov, H. Stögbauer, and P. Grassberger, “Estimating mutual information,” Physical Review E—Statistical, Nonlinear, and Soft Matter Physics **69**, 066138 (2004).
[26]T. Steerneman, “On the total variation and hellinger distance between signed measures; an application to product measures,” Proceedings of the American Mathematical Society **88**, 684–688 (1983).
[27]J. Green and F. Valentine, “On the arzela-ascoli theorem,” Mathematics Magazine **34**, 199–202 (1961).
[28]E. Michael, “Three mapping theorems,” Proceedings of the American Mathematical Society **15**, 410–415 (1964).
[29]E. Giné and J. Zinn, “Bootstrapping general empirical measures,” The Annals of Probability , 851–869 (1990).
[30]G. W. Oehlert, “A note on the delta method,” The American Statistician **46**, 27–29 (1992).
[31]Č. Brukner and A. Zeilinger, “Diffraction of matter waves in space and in time,” Physical Review A **56**, 3804 (1997).

APPENDIX

Probability Space

A *probability space* is the mathematical foundation for modeling a random process. It is formally defined as an ordered triplet

$$\left(\underbrace{\Omega}_{sample\ space}, \underbrace{\mathscr{F}}_{event\ space}, \underbrace{P}_{prob\ measure}\right)$$

that satisfies specific axioms.

The *sample space*, Ω, is a non-empty set containing all possible outcomes of a random experiment. The elements of this set, $\omega \in \Omega$, are called outcomes or sample points. The *event space*, $\mathscr{F}$, or σ-algebra, is a collection of subsets of the sample space Ω. These subsets are called *events*. An event is a set of one or more outcomes. Probabilities of atomic (individual) outcomes may sometimes be interesting, but most of the time, we are interested in computing, estimating, and tracking likelihoods of events. For $\mathscr{F}$ to be a σ-algebra on Ω, it must satisfy the following three axioms:

1. Contains the Sample Space: The sample space Ω must be an element of $\mathscr{F}$, $\Omega \in \mathscr{F}$.
2. Closed under Complementation: If an event A is in $\mathscr{F}$, then its complement, $A^c = \{\omega \in \Omega \mid \omega \notin A\}$, must also be in $\mathscr{F}$. If $A \in \mathscr{F}$, then $A^c \in \mathscr{F}$.
3. Closed under Countable Unions: If $A_1, A_2, A_3, \ldots$ is a countable sequence of events in $\mathscr{F}$, their union must also be in $\mathscr{F}$. If $A_i \in \mathscr{F}$ for all $i = 1, 2, \ldots$, then $\bigcup_{i=1}^{\infty} A_i \in \mathscr{F}$.

For a finite sample space Ω, the event space $\mathscr{F}$ is often taken to be the power set of Ω, which is the set of all possible subsets of Ω.

The *probability measure*, $P : \mathscr{F} \to \mathbb{R}^+_{[0,1]}$, is a function that maps events from the event space to the real numbers. It assigns a specific probability to each event, $P : \mathscr{F} \to [0,1]$. This function must satisfy the *Kolmogorov Axioms*:

1. Non-negativity: The probability of any event must be non-negative. For any event $A \in \mathscr{F}$, $P(A) \geq 0$.
2. Normalization: The probability of the entire sample space (the certain event) is exactly 1, $P(\Omega) = 1$.
3. Countable Additivity (σ-additivity): For any countable collection of pairwise disjoint events $A_1, A_2, A_3, \ldots$ in $\mathscr{F}$ (meaning $A_i \cap A_j = \emptyset$ for $i \neq j$), the probability of their union is equal to the sum of their individual probabilities. * $P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$.

Distribution Identifiability from Modes

Lemma 1: Two distributions $\Phi_1, \Phi_2 \in \mathscr{P}([-\pi, \pi])$ are *identical* if and only if $m_k(\Phi_1) = m_k(\Phi_2)$ for all $k \in \mathbb{Z}$.

Proof of Lemma 1: This follows directly from the uniqueness of Fourier series representation. If $\Phi_1 \neq \Phi_2$, then their difference $\Phi_1 - \Phi_2$ has a non-zero Fourier coefficient, meaning some moment $m_k(\Phi_1) \neq m_k(\Phi_2)$. Conversely, if all moments match, the distributions must be identical almost everywhere. $\square$

Confidence Regions and Uncertainty Quantification

Constructing confidence regions for phase distributions requires adapting standard methods to the manifold structure.

Definition 13 (Circular Confidence Band)**.** A $(1-\alpha)$ simultaneous confidence band for φ_t is a pair of lower and upper band functions $(L_N(\theta), U_N(\theta))$ such that

$$P\left(L_N(\theta) \leq \varphi_t(\theta) \leq U_N(\theta) \text{ for all } \theta \in \mathbb{S}^1\right) \geq 1 - \alpha. \tag{6}$$

Theorem 16 (Bootstrap Confidence Bands)**.** *Let $\widehat{\varphi}_N(\theta)$ be a kernel density estimator on $\mathbb{S}^1$ based on observations $\{\theta_1, \ldots, \theta_N\}$. The bootstrap confidence band based on B bootstrap samples*

$$L_N(\theta) = \widehat{\varphi}_N(\theta) - q^*_{1-\alpha}(\theta),$$

$$U_N(\theta) = \widehat{\varphi}_N(\theta) + q^*_{1-\alpha}(\theta),$$

where $q^*_{1-\alpha}(\theta)$ is the $(1-\alpha)$-quantile of $|\widehat{\varphi}^*_N(\theta) - \widehat{\varphi}_N(\theta)|$ across bootstrap samples, achieves uniform coverage $P(L_N(\theta) \leq \varphi(\theta) \leq U_N(\theta)$ for all $\theta \in \mathbb{S}^1) = 1 - \alpha + o(1)$ as $N \to \infty$.

Proof. Let's start the kernel density estimator

$$\widehat{\varphi}_N(\theta) = \frac{1}{Nh} \sum_{i=1}^{N} K\left(\frac{d_{\mathbb{S}^1}(\theta, \theta_i)}{h}\right), \tag{7}$$

where K is a circular kernel, $h = h_N$ is the bandwidth with $h \to 0$ and $Nh \to \infty$.

Define the centered and scaled process $Z_N(\theta) = \sqrt{Nh}\left(\widehat{\varphi}_N(\theta) - \mathbb{E}[\widehat{\varphi}_N(\theta)]\right)$. For the bootstrap, let $\{\theta^*_1, \ldots, \theta^*_N\}$ be sampled with replacement from $\{\theta_1, \ldots, \theta_N\}$, yielding

$$\widehat{\varphi}^*_N(\theta) = \frac{1}{Nh} \sum_{i=1}^{N} K\left(\frac{d_{\mathbb{S}^1}(\theta, \theta^*_i)}{h}\right). \tag{8}$$

And define the bootstrap process $Z^*_N(\theta) = \sqrt{Nh}\left(\widehat{\varphi}^*_N(\theta) - \widehat{\varphi}_N(\theta)\right)$.

To establish weak convergence of the original process we show that Z_N converges weakly to a Gaussian process. By the central limit theorem for kernel density estimators on manifolds. By pointwise convergence, given $\theta \in \mathbb{S}^1$, $Z_N(\theta) \xrightarrow{d} \mathcal{N}(0, \sigma^2(\theta))$, where $\sigma^2(\theta) = \varphi(\theta) \int K^2(u)du$. We verify tightness using the moment condition. For $\theta_1, \theta_2 \in \mathbb{S}^1$

$$\mathbb{E}|Z_N(\theta_1) - Z_N(\theta_2)|^2 = Nh \cdot \mathrm{Var}\left(\frac{1}{Nh} \sum_{i=1}^{N} \left[K\left(\frac{d_{\mathbb{S}^1}(\theta_1, \theta_i)}{h}\right) - K\left(\frac{d_{\mathbb{S}^1}(\theta_2, \theta_i)}{h}\right)\right]\right) \leq C \cdot \frac{d_{\mathbb{S}^1}(\theta_1, \theta_2)^\gamma}{h^\gamma}, \tag{9}$$

for some $\gamma > 0$ when K is Hölder continuous. Since $\mathbb{S}^1$ is compact, by the Arzelà-Ascoli theorem[27], the process $\{Z_N\}$ is tight in $C(\mathbb{S}^1)$, the space of continuous functions on $\mathbb{S}^1$.

For any finite set $\{\theta_1, \ldots, \theta_k\} \subset \mathbb{S}^1$, $(Z_N(\theta_1), \ldots, Z_N(\theta_k)) \xrightarrow{d} \mathcal{N}(0, \Sigma)$, where $\Sigma_{ij} = \mathrm{Cov}(Z(\theta_i), Z(\theta_j))$ with limiting Gaussian process Z. By the continuous mapping theorem[28], we have $Z_N \xrightarrow{d}$ in $C(\mathbb{S}^1)$, where Z is a centered Gaussian process with covariance function $C(\theta_1, \theta_2) = \lim_{N \to \infty} \mathrm{Cov}(Z_N(\theta_1), Z_N(\theta_2))$.

Next we examine the bootstrap consistency to show that conditionally on the data, Z^*_N converges to the same limit as Z_N. Given the data $\mathscr{D}_N = \{\theta_1, \ldots, \theta_N\}$, the conditional mean and variance are

$$\mathbb{E}^*[Z^*_N(\theta)] = 0, \tag{10}$$

$$\mathrm{Var}^*[Z^*_N(\theta)] = \frac{1}{h} \cdot \frac{1}{N} \sum_{i=1}^{N} K^2\left(\frac{d_{\mathbb{S}^1}(\theta, \theta_i)}{h}\right) - \left(\widehat{\varphi}_N(\theta)\right)^2. \tag{11}$$

As $N \to \infty$ $\mathrm{Var}^*[Z^*_N(\theta)] \xrightarrow{P} \varphi(\theta) \int K^2(u)du = \sigma^2(\theta)$. Te conditional tightness estimates follow similar calculations,

$$\mathbb{E}^*|Z^*_N(\theta_1) - Z^*_N(\theta_2)|^2 \leq C \cdot \frac{d_{\mathbb{S}^1}(\theta_1, \theta_2)^\gamma}{h^\gamma}. \tag{12}$$

Based on[29], we can show conditional weak convergence conditionally on $\mathscr{D}_N$, with probability approaching 1, $Z^*_N \xrightarrow{d^*} Z \quad$ in $C(\mathbb{S}^1)$, where $\xrightarrow{d^*}$ denotes convergence in distribution under the bootstrap measure.

Next we use Hadamard differentiability of the supremum functional $\phi : C(\mathbb{S}^1) \to \mathbb{R}$, where

$$\phi(f) = \sup_{\theta \in \mathbb{S}^1} |f(\theta)|. \tag{13}$$

It's easy to verify that ϕ is Hadamard differentiable at any $f \in C(\mathbb{S}^1)$. For $g \in C(\mathbb{S}^1)$ and $t_n \to 0$

$$\frac{\phi(f + t_n g_n) - \phi(f)}{t_n} \to \phi'_f(g), \tag{14}$$

where the derivative is $\phi'_f(g) = \sup_{\theta \in M(f)} \mathrm{sign}(f(\theta)) \cdot g(\theta)$, with $M(f) = \{\theta : |f(\theta)| = \|f\|_\infty\}$ being the set of maxima.

The Hadamard differentiability follows from this upper bound estimate

$$\left|\sup_\theta |f(\theta) + t_n g_n(\theta)| - \sup_\theta |f(\theta)|\right| \leq \sup_\theta |t_n g_n(\theta)| = |t_n| \|g_n\|_\infty. \tag{15}$$

Since $Z_N \xrightarrow{d} Z$ and ϕ is Hadamard differentiable we can use the functional delta method[30]

$$\phi(Z_N) = \sup_{\theta\in\mathbb{S}^1} |Z_N(\theta)| \xrightarrow{d} \phi(Z) = \sup_{\theta\in\mathbb{S}^1} |Z(\theta)|. \tag{16}$$

Similarly, by bootstrap consistency $\phi(Z_N^*) = \sup_{\theta\in\mathbb{S}^1} |Z_N^*(\theta)| \xrightarrow{d^*} \phi(Z) = \sup_{\theta\in\mathbb{S}^1} |Z(\theta)|$, conditionally on the data, with probability approaching 1.

To establish the coverage probability let $c_{1-\alpha}$ be the $(1-\alpha)$-quantile of $\sup_{\theta\in\mathbb{S}^1} |Z(\theta)|$ and $c^*_{1-\alpha}$ be the bootstrap estimate (the $(1-\alpha)$-quantile of $\sup_{\theta\in\mathbb{S}^1} |Z_N^*(\theta)|$).

By bootstrap consistency we have $c^*_{1-\alpha} \xrightarrow{P} c_{1-\alpha}$ and the coverage probability is

$$P\left(\varphi(\theta) \in \left[\widehat{\varphi}_N(\theta) - c^*_{1-\alpha}/\sqrt{Nh}, \widehat{\varphi}_N(\theta) + c^*_{1-\alpha}/\sqrt{Nh}\right] \text{ for all } \theta\right) \tag{17}$$

$$= P\left(\sup_{\theta\in\mathbb{S}^1} \sqrt{Nh}|\widehat{\varphi}_N(\theta) - \varphi(\theta)| \le c^*_{1-\alpha}\right) = P\left(\sup_{\theta\in\mathbb{S}^1} |Z_N(\theta) + R_N(\theta)| \le c^*_{1-\alpha}\right), \tag{18}$$

where $R_N(\theta) = \sqrt{Nh}(\mathbb{E}[\widehat{\varphi}_N(\theta)] - \varphi(\theta)) = O(\sqrt{Nh}\cdot h^2) = o(1)$ under standard bandwidth conditions.

Therefore,

$$P\left(\sup_{\theta\in\mathbb{S}^1} |Z_N(\theta) + R_N(\theta)| \le c^*_{1-\alpha}\right) = P\left(\sup_{\theta\in\mathbb{S}^1} |Z_N(\theta)| \le c^*_{1-\alpha} + o(1)\right) \tag{19}$$

$$\longrightarrow P\left(\sup_{\theta\in\mathbb{S}^1} |Z(\theta)| \le c_{1-\alpha}\right) = 1-\alpha. \tag{20}$$

The final uniform coverage comes from considering the pointwise quantiles $q^*_{1-\alpha}(\theta)$ that satisfy

$$\sup_{\theta\in\mathbb{S}^1} |q^*_{1-\alpha}(\theta) - c^*_{1-\alpha}| = o_P(1), \tag{21}$$

under smoothness conditions on φ. This implies that

$$P(L_N(\theta) \le \varphi(\theta) \le U_N(\theta) \text{ for all } \theta \in \mathbb{S}^1) = 1-\alpha + o(1). \tag{22}$$

□

Kime representation and Statistical Formulation of Quantum Mechanics

NOTE: This section may eventually transform into a separate, independent and deeper, study of the relation between *kime-representation* and the Moyal's statistical formulation of QM[16].

In 1949, Moyal proposed a statistical formulation of quantum mechanics[16]. This section/paper examines the theoretical foundations, historical context, and potential relations to KPT. Complementing the classical formulation of quantum mchanics as a theory of abstract state vectors, the Moyal interpretation describes quantum mechanics as non-deterministic statistical dynamics on phase space. The latter is a joint probability distribution function $F(p,q)$ of non-commuting observables, called the Wigner quasi-probability distribution, which arises directly from the wave function $\psi(q)$

$$F(p,q) = \frac{1}{\pi\hbar}\int_{-\infty}^{\infty} \psi^*(q-y)\psi(q+y)e^{2ipy/\hbar}dy.$$

In KPT, a longitudinal process is not just a longitudinal time-course, $y(t)$, but as a kime-surface $\mathscr{S}(t,\theta)$ over the kime-domain $te^{i\theta}$. The phase θ is a latent variable whose distribution $\varphi_t(\theta)$ must be inferred. This is directly analogous to Moyal's phase space (q,p). In KPT, the time t is the classical, directly observable variable, like the position q, while the phase θ is a conjugate, latent variable, like the classical momentum p, that modulates the repeatedly observed longitudinal process. The kime-surface $\mathscr{S}(t,\theta)$ is the functional equivalent of Moyal's $F(p,q)$, a joint representation of the system in its complete kime state space. The synergy between Moyer's and KPT representations is the lift of the system into a higher-dimensional, conjugate space, Moyal phase space or KPT kime-phase.

Moyal showed that the non-commutativity of classical quantum operators, e.g., $[\hat{q},\hat{p}] = i\hbar$, is encoded in the structure of the phase-space distribution. The differential operator known as *Moyal bracket* governs the time evolution of $F(p,q)$ and reduces to the classical Poisson bracket in the limit $\hbar \to 0$.

Moyal bracket

Moyal derived the equation of motion for the Wigner quasi-probability distribution $F(p,q;t)$ and showed that it has a specific Moyal bracket form, a deformed classical Poisson bracket. Given a wave function $\psi(q)$ (pure state $\rho = |\psi\rangle\langle\psi|$), the phase-space distribution is

$$F(p,q) = \frac{1}{\pi\hbar}\int_{-\infty}^{\infty} \psi^*(q-y)\psi(q+y)e^{2ipy/\hbar}dy.$$

This function $F(p,q)$ is real-valued but can be negative, i.e., it's a *quasi-probability*. It encodes all the information in the wave function $\psi(q)$. To find how $F(p,q;t)$ evolves in time, the quantum Liouville equation, Moyal Equation, is derived from the Schrödinger equation

$$i\hbar\frac{\partial\psi}{\partial t} = \hat{H}\psi,$$

where $\hat{H}$ is the Hamiltonian operator, typically $\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{q})$. Assuming a real potential $V(q)$, so that $\hat{H}$ is Hermitian operator, the complex conjugate equation is

$$-i\hbar\frac{\partial\psi^*}{\partial t} = \hat{H}\psi^*.$$

Taking the time derivative of $F(p,q;t)$ we obtain

$$\frac{\partial F}{\partial t} = \frac{1}{\pi\hbar}\int_{-\infty}^{\infty}\left[\frac{\partial\psi^*}{\partial t}(q-y)\psi(q+y) + \psi^*(q-y)\frac{\partial\psi}{\partial t}(q+y)\right]e^{2ipy/\hbar}dy$$

and substituting the Schrödinger equation yields

$$\frac{\partial F}{\partial t} = \frac{1}{\pi\hbar}\int_{-\infty}^{\infty}\left[\left(\frac{i}{\hbar}\hat{H}\psi^*\right)(q-y)\psi(q+y) + \psi^*(q-y)\left(-\frac{i}{\hbar}\hat{H}\psi\right)(q+y)\right]e^{2ipy/\hbar}dy.$$

This expression simplifies to

$$\frac{\partial F}{\partial t} = \frac{i}{\hbar}\cdot\frac{1}{\pi\hbar}\int_{-\infty}^{\infty}\left[\left(\hat{H}\psi^*\right)(q-y)\psi(q+y) - \psi^*(q-y)\left(\hat{H}\psi\right)(q+y)\right]e^{2ipy/\hbar}dy. \tag{23}$$

This equation 23 may also be written as a differential operator acting on $F(p,q)$ and the classical Hamiltonian $H(p,q)$. The term in brackets corresponds to a *star product*, aka *Moyal product*, $A \star B$, of two phase-space functions $A(p,q)$ and $B(p,q)$ is defined by

$$(A\star B)(p,q) = A(p,q)\exp\left[\frac{i\hbar}{2}\left(\frac{\overleftarrow{\partial}}{\partial q}\frac{\overrightarrow{\partial}}{\partial p} - \frac{\overleftarrow{\partial}}{\partial p}\frac{\overrightarrow{\partial}}{\partial q}\right)\right]B(p,q), \tag{24}$$

which is an infinite-order differential operator. The arrows indicate that the derivatives act on the function to the left, $\overleftarrow{\partial}$, or to the right, $\overrightarrow{\partial}$. As a result, the time evolution is expressed as

$$\frac{\partial F}{\partial t} = \frac{i}{\hbar}\left(H\star F - F\star H\right),$$

where the quantity $\frac{i}{\hbar}(H\star F - F\star H)$ is the *Moyal bracket*, $\{H,F\}_M$,

$$\{H,F\}_M := \frac{i}{\hbar}(H\star F - F\star H).$$

Hence, the fundamental equation of motion is

$$\boxed{\frac{\partial F}{\partial t} = \{H,F\}_M},$$

which is the quantum Liouville equation, or the *Moyal equation* governing the deterministic evolution of the quasi-probability distribution $F(p,q;t)$ in the phase space.

To show that the Moyal bracket is a deformation of the classical Poisson bracket, we expand the star product in powers of $\hbar$. The exponential in the star product has the following Taylor series expansion

$$\exp\left[\frac{i\hbar}{2}\left(\overleftarrow{\partial_q}\overrightarrow{\partial_p}-\overleftarrow{\partial_p}\overrightarrow{\partial_q}\right)\right]=1+\frac{i\hbar}{2}\left(\overleftarrow{\partial_q}\overrightarrow{\partial_p}-\overleftarrow{\partial_p}\overrightarrow{\partial_q}\right)+\frac{1}{2!}\left(\frac{i\hbar}{2}\right)^2\left(\overleftarrow{\partial_q}\overrightarrow{\partial_p}-\overleftarrow{\partial_p}\overrightarrow{\partial_q}\right)^2+\underbrace{\cdots}_{\text{higher order terms}}. \tag{25}$$

When computing $H\star F-F\star H$, the leading term of order $\hbar^1$ cancels out because $H\cdot F-F\cdot H=0$. The next term is

$$H\star F-F\star H=\left[HF+\frac{i\hbar}{2}\left(\frac{\partial H}{\partial q}\frac{\partial F}{\partial p}-\frac{\partial H}{\partial p}\frac{\partial F}{\partial q}\right)+O(\hbar^2)\right]-\left[FH+\frac{i\hbar}{2}\left(\frac{\partial F}{\partial q}\frac{\partial H}{\partial p}-\frac{\partial F}{\partial p}\frac{\partial H}{\partial q}\right)+O(\hbar^2)\right]$$

can be simplified to

$$H\star F-F\star H=\frac{i\hbar}{2}\left(\frac{\partial H}{\partial q}\frac{\partial F}{\partial p}-\frac{\partial H}{\partial p}\frac{\partial F}{\partial q}\right)-\frac{i\hbar}{2}\left(\frac{\partial F}{\partial q}\frac{\partial H}{\partial p}-\frac{\partial F}{\partial p}\frac{\partial H}{\partial q}\right)+O(\hbar^2)$$

$$=\frac{i\hbar}{2}\left(2\frac{\partial H}{\partial q}\frac{\partial F}{\partial p}-2\frac{\partial H}{\partial p}\frac{\partial F}{\partial q}\right)+O(\hbar^2)$$

$$=i\hbar\left(\frac{\partial H}{\partial q}\frac{\partial F}{\partial p}-\frac{\partial H}{\partial p}\frac{\partial F}{\partial q}\right)+O(\hbar^2).$$

Plugging this into the Moyal bracket, equ. 25, yields

$$\{H,F\}_M=\frac{i}{\hbar}(H\star F-F\star H)=\frac{i}{\hbar}\left[i\hbar\left(\frac{\partial H}{\partial q}\frac{\partial F}{\partial p}-\frac{\partial H}{\partial p}\frac{\partial F}{\partial q}\right)+O(\hbar^2)\right]$$

$$=-\left(\frac{\partial H}{\partial p}\frac{\partial F}{\partial q}-\frac{\partial H}{\partial q}\frac{\partial F}{\partial p}\right)+O(\hbar),$$

where the term in parentheses is precisely the definition of the classical Poisson bracket $\{H,F\}_{PB}$

$$\{H,F\}_{PB}=\frac{\partial H}{\partial q}\frac{\partial F}{\partial p}-\frac{\partial H}{\partial p}\frac{\partial F}{\partial q}.$$

Therefore, $\{H,F\}_M=\{H,F\}_{PB}+O(\hbar)$, and the equation of motion becomes

$$\frac{\partial F}{\partial t}=\{H,F\}_M=\{H,F\}_{PB}+O(\hbar).$$

Asymptotically, as $\hbar\to 0$, the higher-order terms vanish, and we recover the classical Liouville equation

$$\lim_{\hbar\to 0}\frac{\partial F}{\partial t}=\{H,F\}_{PB},$$

which governs the evolution of a classical probability density $F(p,q;t)$ in phase space under Hamiltonian dynamics.

In essence, quantum dynamics on phase space via the Moyal equation is a direct order-by-order deformation of classical Hamiltonian dynamics with $\hbar$ as the deformation parameter. This observation links classical and quantum statistical mechanics.

Statistical Physical Interpretation of Commutation Relations

Section II of this KPT paper established the canonical commutation relations $[\hat{T},\hat{P}_t]=i\hbar_\kappa\mathbf{I}$ and $[\hat{\Theta},\hat{L}_\theta]=i\hbar_\kappa\mathbf{I}$. These are the exact analogs of the position-momentum commutator in quantum mechanics. The KPT operators $\hat{T}$ (time multiplication) and $\hat{P}_t$ (time differentiation) are conjugate, just as $\hat{q}$ and $\hat{p}$ are.

Moyal's work connects the physical and statistical interpretations of these commutation relations and confirms that the non-commutativity is not an artifact but a signature of the underlying statistical structure. Moyal's derivation of the dynamical equation for $F(p,q)$ (his Eq. 7.8) resembles how KPT's kime-surface dynamics could be formalized for non-stationary processes, which potentially may lead to a similar Moyal equation for kime-surfaces.

A major focus of Moyal's paper is the issue that his phase-space distribution $F(p,q)$ can take *negative values*, and thus cannot be interpreted as a true probability. This feature may be inconvenient, but not necessarily a flaw, but a feature – $F(p,q)$ is a *quasi-probability* that encodes *quantum interference*. True probabilities are obtained by integrating over one variable, e.g., $\int F(p,q)dp = |\psi(q)|^2$, or by forming non-negative linear combinations, such as mixed states.

In KPT, the latent phase θ is not directly observable, just as p is not simultaneously observable with q in quantum mechanics. The distribution $\varphi_t(\theta)$ is inferred, not measured. While $\varphi_t(\theta)$ is constrained to be non-negative (a true probability), the kime-surface $\mathscr{S}(t,\theta)$ is not. It can be negative or oscillatory, encoding the interference between different phase states, just like the Wigner function. Potential negativity or non-classical behavior of the kime-surface $\mathscr{S}(t,\theta)$ may be a source of information, rather than a nuisance. This reflects the coherent, wave-like nature of the underlying process. KPT's regularization, e.g., via the Sobolev penalty λR_p, can manage this quantumness for practical estimation, which may be viewed as fundamental.

Moyal explicitly frames quantum dynamics as a dynamical stochastic process that is more general than classical (deterministic) statistical mechanics. A derivate characteristic function $L(\tau,\theta|p,q)$ describes the infinitesimal evolution of the phase-space distribution, leading to an integro-differential master equation. Similarly, KPT currently focuses on the static or quasi-static recovery of $\varphi_t(\theta)$ at each time point t. The framework treats time t as a parameter, not a dynamic variable. Moyal's approach resembled the KPT dynamic modeling of the phase distribution itself. It may be feasible to define a kime-space master equation for $\partial\varphi_t(\theta)/\partial t$, governed by a kime-analog of Moyal's derivate moments. Instead of estimating $\varphi_t(\theta)$ independently at each t, one could impose a dynamical prior or constraint based on a stochastic differential equation on the circle $\mathbb{S}^1$. This would lead to a more effective and temporally coherent estimation of the phase process, especially for sparse or noisy data.

On the one hand, Moyal's strategy provides the foundational physics and mathematics for representing systems with conjugate, non-commuting variables in a statistical phase space. On the other hand, KPT provides a pragmatic statistical and computational representation of repeated-measurement longitudinal data in complex systems (biological, physical, etc.). Moyal's formulation is for fundamental particles, whereas KPT deals with complex systems such as brains, fluids, and materials. Both approaches are highly synergistic, seeking to recover a latent, conjugate variable, e.g., momentum p or phase θ, that modulates the observable, position q or signal $y(t)$.

Bessel-Fourier Basis

The Bessel-Fourier functional basis is used to represent solutions to problems with circular (phase) symmetry, e.g., solving partial differential equations like the wave or heat equation in cylindrical coordinates. It combines Bessel functions (radial part) and Fourier modes (angular part). In polar (kime) coordinates $\kappa = (r,\theta)$, the basis functions are

$$\psi_{nm}(r,\theta) = J_n(k_{nm}r)e^{in\theta}, \quad n \in \mathbb{Z}, \quad m = 1,2,\ldots,$$

where J_n is the Bessel function of the first kind of order n and k_{nm} is the m-th root of $J_n(ka) = 0$ for a disk of radius a (Dirichlet boundary condition).

The Bessel-Fourier functions represent an orthogonal and complete basis over square-integrable kime functions, over a kime disk,

$$\int_0^a \int_0^{2\pi} \psi_{nm}(r,\theta)\psi^*_{n'm'}(r,\theta)rdrd\theta = \delta_{nn'}\delta_{mm'} \cdot \frac{\pi a^2}{2}[J_{n+1}(k_{nm}a)]^2.$$

For instance, a function kime $f(r,\theta)$ can be expanded as

$$f(r,\theta) = \sum_{n=-\infty}^{\infty}\sum_{m=1}^{\infty} c_{nm}J_n(k_{nm}r)e^{in\theta},$$

where the coefficients are defined by

$$c_{nm} = \frac{2}{\pi a^2[J_{n+1}(k_{nm}a)]^2}\int_0^a \int_0^{2\pi} f(r,\theta)J_n(k_{nm}r)e^{-in\theta}rdrd\theta.$$

the Bessel-Fourier basis is applicable for solving the Helmholtz equation in cylindrical domains, analyzing vibrations of circular membranes (e.g., drums), and studying electromagnetic fields in waveguides. More generally, this representation leverages the orthogonality of both Bessel functions and Fourier exponentials, making it ideal for problems with circular symmetry.

Integral Probability Metric (IPM) and Cramér-von Mises (CvM) criterion

For high-dimensional data, integral probability metrics (IPMs), such as the Sobolev CvM, may be used to compare two probability distributions, particularly. The Sobolev-CvM metric combines the principles of Sobolev spaces from functional analysis with the statistical Cramér-von Mises (CvM) criterion. The Sobolev-CvM measures the distance between two distributions, P and Q, by comparing their *mean discrepancy* over a class of functions known as a Sobolev ball. This process involves the following.

1. Select a function class: A "critic" function, f, is chosen from a Sobolev ball. A Sobolev space is a function space that includes functions and their weak derivatives, making them regular and smooth. The Sobolev ball is the set of all such functions with a specific norm bound.
2. Compare mean values: The mean value of the critic function f is calculated under each distribution, $\mathbb{E}_{x\sim P}[f(x)]$ and $\mathbb{E}_{x\sim Q}[f(x)]$.
3. Take the supremum: The metric is defined as the maximum difference between these mean values over all possible critic functions in the Sobolev ball.

The Sobolev CvM metric is defined by

$$d(P,Q) = \sup_{f\in\mathscr{S}} |\mathbb{E}_{x\sim P}[f(x)] - \mathbb{E}_{x\sim Q}[f(x)]|,$$

where $\mathscr{S}$ is the unit ball in a specific Sobolev space. The connection to the traditional Cramér-von Mises test lies in how the Sobolev IPM can be interpreted. In a simpler, one-dimensional case, a Sobolev IPM can be seen as an extension of the standard Cramér-von Mises statistic to higher dimensions. The standard Cramér-von Mises test is a statistical goodness-of-fit test that measures the difference between an empirical cumulative distribution function (CDF) and a theoretical CDF using the sum of squared differences.

The Sobolev-CvM generalizes the classical Cramér-von Mises test by measuring the discrepancy between two distributions in a high-dimensional space. It achieves this by comparing conditional CDFs along each dimension, weighted by a dominant measure. Sobolev-CvM has applications in machine learning. For instance, it can be used to train generative adversarial networks (GANs), which learn to create new data instances with the same characteristics as a given training set. The Sobolev-CvM's inherent ability to enforce smoothness on the critic function makes it effective in semi-supervised learning tasks. Also, Sobolev-CvM can be used in text generation tasks by exploiting the intrinsic conditioning implied by the metric.

Estimating the *de-biased* phase, $\widehat{\varphi}_t(\cdot)$, and its variance

Here we explicate the *de-biased* $\widehat{\varphi}_t(\cdot)$ estimates and their variance using the stored spectral mixed-moment arrays $\{\widehat{M}(\omega,t)\}$. The notation and FFT conventions match the formulation in Sections III and IV, including $\Lambda_p(\omega)$, Wiener-Sobolev filtering, and the posterior Fourier statistics.

This R/Rmd implementation makes the following *assumptions:*

1. For each time t, we have porecomputed the vectors $\widehat{M}(\omega,t) \in \mathbb{C}^M$ and $\widehat{F}(\omega,t) \in \mathbb{C}^M$ on the $(2J+1)$-grid $\omega_k = \frac{2\pi k}{M}$, $k = -J,\ldots,J$, with $M = 2J+1$.
2. For variance, we either have subject-level contributions $\{\widehat{M}_j(\omega,t)\}_{j=1}^N$ (recommended), where $\widehat{M}(\omega,t) = \frac{1}{N}\sum_j \widehat{M}_j(\omega,t)$, or alternatively, use a bootstrap (requires $y_{j,k}$ and the E-step).
3. FFT convention: forward FFT has no scale; IFFT has $1/M$ scale (common in libraries).
4. All phase integrals use $d\theta/(2\pi)$.

R/Rmd Code Example: (Estimating the de-biased $\widehat{\varphi}_t(\theta)$ and its variance)

Below is an R/Rmd implementation of the de-biased estimation algorithm including *variance estimation*, using the spectral delta method, the *Rayleigh harmonic test* for phase concentration, and *error handling* and numerical stability.

```
```{r de-biasedEstim}
#' Debiased Phase Density Estimation via Spectral Deconvolution
#'
#' @description Implements the debiased KPT estimator for the phase density $\varphi_t(\theta)$
```
```

```
#' using spectral deconvolution with Sobolev regularization.
#'
#' @param Mhat complex length-M vector: stored mixed-moment spectrum $\widehat{M}(\omega,t)$
#' @param Fhat complex length-M vector: stored surface spectrum $\widehat{F}(\omega,t)$
#' @param lambda_phi phase-penalty weight $\lambda_{\Phi} \geq 0$ used in KPT
#' @param Lambda_p Sobolev weights $\Lambda_p(\omega) = (2\sin(\omega/2))^{2p}$
#' on the same grid
#' @param Mhat_by_subject optional complex matrix N×M with rows $\widehat{M}_j(\omega,t)$
#' for variance estimation
#' @param ridge_tau small positive ridge for numerical stability
#' (e.g., $10^{-10} \cdot \text{median}(|\widehat{F}|^2)$)
#'
#' @return list with components:
#' \item{phi_debias_theta}{debiased phase density on $\theta$-grid, length M
#'  (nonnegative, sums to 1)}
#' \item{var_phi_debias_theta}{variance at each $\theta$ (if subject-level spectra available)}
#' \item{varPhi_debias}{variance in spectral domain (if subject-level spectra available)}
#'
#' @details
#' The debiased estimator is defined as:
#' \[
#' \widetilde{\Phi}(\omega) = \frac{\overline{\widehat{F}(\omega)}}
#' {|\widehat{F}(\omega)|^2} \widehat{M}(\omega)
#' \]
#' with IFFT transformation to the time domain and simplex projection.
#'
#' @examples
#' # Example usage with simulated data
#' M <- 101; J <- (M-1)/2
#' omega <- 2*pi * (-J:J)/M
#'
#' # Simulate spectra
#' Fhat <- exp(-0.5*omega^2) + 0.1i*omega
#' Mhat <- Fhat * (1 + 0.2*exp(1i*omega))  # Convolution relation
#' lambda_phi <- 0.1
#' Lambda_p <- (2*sin(omega/2))^4  # p=2 Sobolev penalty
#'
#' result <- debiased_phase_density(Mhat, Fhat, lambda_phi, Lambda_p)
#' plot(result$phi_debias_theta, type = "l", main = "Debiased Phase Density")
debiased_phase_density <- function(Mhat, Fhat, lambda_phi, Lambda_p,
                                  Mhat_by_subject = NULL, ridge_tau = 1e-10) {

  # Constants
  M <- length(Mhat)
  J <- (M - 1)/2
  omega <- 2*pi * (-J:J)/M  # Frequency grid: _k = 2k/M, k = -J,...,J

  # --- Step 0: Precompute filters ---
  # H() = |F()|² / (|F()|² + _ _p()) - shrinkage filter used by KPT M-step
  Fhat_sq <- Mod(Fhat)^2
  H <- Fhat_sq / (Fhat_sq + lambda_phi * Lambda_p)

  # L0() = conj(F()) / max(|F()|², ridge_tau) - debiasing filter
  denominator <- pmax(Fhat_sq, ridge_tau)
  L0 <- Conj(Fhat) / denominator

  # --- Step 1: Debiased spectral estimate of  ---
  # () = L0() · M() = _KPT()/H()
```

```
Phi_tilde <- L0 * Mhat

# --- Step 2: Inverse FFT to -domain ---
# Uses 1/M scaling as required by the algorithm
phi_debias_theta <- fft(Phi_tilde, inverse = TRUE) / M

# Ensure result is real-valued (should be due to symmetry)
phi_debias_theta <- Re(phi_debias_theta)

# --- Step 3: Project to probability simplex ---
# Nonnegativity constraint
phi_debias_theta <- pmax(phi_debias_theta, 0)

# Unit mass constraint: (1/M) _ (_) = 1   () d/(2)
scale <- sum(phi_debias_theta) / M

if (scale == 0) {
  # Fallback to uniform distribution
  phi_debias_theta <- rep(1/M, M)
} else {
  phi_debias_theta <- phi_debias_theta / scale
}

# --- Step 4: Global phase anchoring (remove rotation ambiguity) ---
# Force arg (1) = 0 by circular shift (consistent with Section 3 convention)
# Implementation: rotate Phi_tilde by appropriate phase
phase_shift <- -Arg(phi_debias_theta[1])
phi_debias_theta <- phi_debias_theta * exp(1i * phase_shift)
phi_debias_theta <- Re(phi_debias_theta)  # Should remain real

# --- Step 5: Variance estimation (if subject-level data available) ---
var_phi_debias_theta <- NULL
varPhi_debias <- NULL

if (!is.null(Mhat_by_subject)) {
  N <- nrow(Mhat_by_subject)

  # --- 5a: Sample covariance of subject-level spectra ---
  Z_bar <- colMeans(Mhat_by_subject)  # Should equal Mhat

  # Compute sample covariance matrix S_Z[u, v]
  S_Z <- matrix(0, M, M)
  for (j in 1:N) {
    residuals <- Mhat_by_subject[j,] - Z_bar
    S_Z <- S_Z + outer(residuals, Conj(residuals))
  }
  S_Z <- S_Z / (N - 1)

  # Covariance of the mean M is S_Z / N
  Cov_M <- S_Z / N

  # --- 5b: Transport covariance through debiasing map ---
  # Cov_ = diag(L0) · Cov_M · diag(conj(L0))
  L0_matrix <- diag(L0)
  L0_conj_matrix <- diag(Conj(L0))
  Cov_Phi_tilde <- L0_matrix %*% Cov_M %*% L0_conj_matrix

  # --- 5c: Spectral variances (diagonal elements) ---
```

```
    varPhi_debias <- diag(Cov_Phi_tilde)

    # --- 5d: -domain variances via IFFT ---
    theta_grid <- 2*pi * (0:(M-1))/M  # _ = 2(-1)/M

    var_phi_debias_theta <- numeric(M)
    for (ell in 1:M) {
      # v[u] = (1/M) exp(i u _)
      v <- (1/M) * exp(1i * outer(omega, theta_grid[ell]))

      # var = v^H · Cov_ · v (take real part)
      var_phi_debias_theta[ell] <- Re(Conj(t(v)) %*% Cov_Phi_tilde %*% v)
    }
  }

  # Return results
  list(
    phi_debias_theta = phi_debias_theta,
    var_phi_debias_theta = var_phi_debias_theta,
    varPhi_debias = varPhi_debias,
    Phi_tilde = Phi_tilde,  # Optional: return spectral estimate
    H = H,                  # Optional: return shrinkage filter
    L0 = L0                 # Optional: return debiasing filter
  )
}

#' Rayleigh Harmonic Test for Phase Concentration
#'
#' @description Implements the Rayleigh test using the debiased spectral estimate
#' at the fundamental frequency  = 2/M.
#'
#' @param Phi_tilde debiased spectral estimate $\widetilde{\Phi}(\omega)$
#' @param varPhi_debias variance of $\widetilde{\Phi}(\omega)$ (optional)
#' @param omega frequency grid
#'
#' @return list with test statistics and p-values
#'
#' @details
#' Under the null hypothesis of uniform phase distribution, the test statistic
#' follows a $\chi^2_2$ distribution:
#' \[
#' R = \frac{|\widetilde{\Phi}(\omega_1)|^2}{\text{Var}
#' [\widetilde{\Phi}(\omega_1)]} \sim \chi^2_2
#' \]
rayleigh_harmonic_test <- function(Phi_tilde, varPhi_debias = NULL, omega) {
  # Find fundamental frequency  (closest to 2/M)
  M <- length(omega)
  omega1 <- 2*pi/M
  idx <- which.min(abs(omega - omega1))

  # Extract harmonic component
  Phi1 <- Phi_tilde[idx]

  if (is.null(varPhi_debias)) {
    # Without variance, return only the magnitude
    return(list(
      magnitude = Mod(Phi1),
      phase = Arg(Phi1),
```

```
      warning = "Variance not available for formal test"
    ))
  }

  # Rayleigh test statistic
  var_Phi1 <- varPhi_debias[idx]
  if (var_Phi1 > 0) {
    R <- Mod(Phi1)^2 / var_Phi1
    p_value <- pchisq(R, df = 2, lower.tail = FALSE)
  } else {
    R <- Inf
    p_value <- 0
  }

  list(
    magnitude = Mod(Phi1),
    phase = Arg(Phi1),
    test_statistic = R,
    p_value = p_value,
    frequency_index = idx,
    fundamental_frequency = omega1
  )
}

# --- Example usage  ---
#'
#' ## Mathematical Background
#'
#' The debiased estimator is derived from the KPT (Kernel Phase Tracking) framework:
#'
#' **KPT Estimator:**
#' \[
#' \widehat{\Phi}_{\text{KPT}}(\omega) = \frac{\overline{\widehat{F}(\omega)}}
#' {|\widehat{F}(\omega)|^2 + \lambda_\Phi \Lambda_p(\omega)} \widehat{M}(\omega)
#' \]
#'
#' **Debiased Version:**
#' \[
#' \widetilde{\Phi}(\omega) = \frac{\widehat{\Phi}_{\text{KPT}}(\omega)}{H(\omega)} =
#' \frac{\overline{\widehat{F}(\omega)}}{|\widehat{F}(\omega)|^2} \widehat{M}(\omega)
#' \]
#' where $H(\omega) = \frac{|\widehat{F}(\omega)|^2}{|\widehat{F}(\omega)|^2
#' + \lambda_\Phi \Lambda_p(\omega)}$ is the shrinkage factor.
#'
#' The phase density is obtained via inverse Fourier transform:
#' \[
#' \widetilde{\varphi}_t(\theta) =
#' \frac{1}{2\pi} \sum_{\omega} \widetilde{\Phi}(\omega) e^{i\omega\theta}
#' \]
#' with simplex projection to ensure $\widetilde{\varphi}_t(\theta) \geq 0$
#' and $\int \widetilde{\varphi}_t(\theta) d\theta = 1$.
```

Direct Linkage between Observed Double-Slit Experiment Data and Kime-Surface Model

This section explicates the mathematical–physics formulation tying the *kime objects*

$$\left\{ \underbrace{S(t,\theta)}_{kime-surface}, \underbrace{\varphi_t(\theta)}_{phase-law} \right\}$$

to the *1-D position distribution* actually measured on the screen in the double-slit experiment. We reflect on the KPT-added value even when the emission order is exchangeable, when the double-slit experimental conditions are strictly fixed, N repeated runs each with k particle emissions and detections per run is expected to yield exactly the same interference distribution as a single experimental run with $N \times k$ particle emissions.

This experiment simulates screen positions, uses the KPT reconstructions to predict the position density $p(x)$, and produces an interactive plot with open-circle screen hits against KPT predictive curves (with proper anchoring, axis orientation, and uncertainty).

We will show how to transforming kime (t,θ) to screen position x predictions using Fraunhofer regime (far field)[31] slit separation d, slit width α, screen distance L, and wavelength λ, the screen intensity at horizontal coordinate x is

$$\underbrace{I(x;\mu)}_{\text{screen intensity}} = I_0 \underbrace{\text{sinc}^2\left(\frac{\pi a}{\lambda L}, x\right)}_{\text{envelope } E(x)} \underbrace{\left[1+\cos\left(\alpha x+\mu\right)\right]}_{\text{interference}}, \qquad \alpha \equiv \frac{2\pi d}{\lambda L}.$$

A small path-difference (or source) phase offset μ simply shifts the fringes. The detection density on the screen is proportional to $I(x;\mu)$.

Let's define the phase coordinate $\theta(x) = \alpha x$ and write the time-aggregated kime-surface as a 2π-periodic function

$$\bar{S}(\theta) \approx B + A_1 \cos\theta + A_2 \cos(2\theta) + \cdots .$$

Then, the time-aggregated KPT phase law is $\bar{\varphi}(\phi)$ on $\phi \in [-\pi,\pi)$, both are *anchored* as described below, and the KPT prediction for the exchangeable double-slit position density is

$$\boxed{\underbrace{p_{\text{KPT}}(x)}_{\substack{\text{KPT prediction}\\ \text{of position density}}} \propto E(x)\left(\bar{S}*\bar{\varphi}\right)\left(\theta(x)\right)}.$$

Thus, the KPT prediction of the position density, $p_{\text{KPT}}(x)$, is expressed as a product of the *envelope* $E(x)$ and a circular convolution of the time-aggregated kime-surface with the phase distribution, $\bar{S}*\bar{\varphi}$, which is evaluated at $\theta = \alpha x$. This directly connects *2-D kime objects* to the 1-D measured position quantity. When the double-slit experimental run conditions are fixed (exchangeable emissions), the time index is not causal. Thus, we simply use the time-aggregated kime-surface and the local phase average

$$\bar{S}(\theta) = \frac{1}{K}\sum_k S(t_k,\theta), \qquad \bar{\varphi}(\theta) = \frac{1}{K}\sum_k \varphi_{t_k}(\theta).$$

If μ varies across runs (e.g., mechanical jitter, air currents, micro-drift), then the *fringe shift* distribution is exactly a phase law φ. KPT recovers $\bar{\varphi}$ (shape and concentration) and a debiased $\bar{S}$ (the interference structure). Together they give calibrated predictive density for screen hits x, uncertainty quantification, i.e., predictive intervals, and a Bayesian encoder for latent physical offsets μ, i,e, the posterior $p(\mu \mid x_{1:k})$ with KPT prior.

Position-density prediction is obtained using the fitted S,φ (kime-model), by computing the time-averaged $\bar{S},\bar{\varphi}$ and the KPT prediction of the position density $p_{\text{KPT}}(x)$, and comparing with the empirical histogram of hits on the detection screen. In addition to the visual KPT model exploration, we quantify the KPT model performance using the log predictive density and the *continuous ranked probability score (CRPS)*, which are metrics used to assess the accuracy of probabilistic forecasts.

One can also demonstrate a run-wise phase-offset inversion treating a run's offset μ as latent. Using the time-averaged KPT model $\bar{S}$ and the recovered phase law $\bar{\varphi}$ as a prior on μ, we will compute the *posterior predictive density*

$$p(\mu \mid x_{1:k}) \propto \bar{\varphi}(\mu)\prod_i I(x_i;\mu),$$

where $I(x_i;\mu) \propto E(x_i)[\bar{S}(\alpha x_i + \mu)])$.

This shows how KPT converts repeated observations into physical parameter inference, and allows us to answer questions such as *What phase shift (or source offset) explains the current partial run?*

For rotation invariance, we anchor both estimates before mapping to the position x.

Diagnostic plots allow us to examine the proper scoring for positions, using the log-predictive density score and CRPS metrics.

KPT Model Results Interpretation:

Predictive accuracy (x-space). Both algorithms appear well-calibrated for the screen-position density $p(x)$.

MLPD: $\text{MLPD}_{\text{FFT}} = 5.1230$ vs. $\text{MLPD}_{\text{GEM}} = 5.0936$. This implies that FFT is slightly better, higher ↑MLPD is better.

CRPS: $\text{CRPS}_{\text{FFT}} = 5.51 \times 10^{-6}$ vs. $\text{CRPS}_{\text{GEM}} = 1.55 \times 10^{-5}$, which again suggests that FFT better, lower ↓CRPS is better.

The absolute scale of these metrics depends on the x-units and grid, but the ranking is the key. After making the KPT prediction pipeline physically valid (non-negative intensity with the same rectification used in the simulator), both KPT-GEM and KPT-FFT produce very similar, high-quality $p(x)$ models, with a small advantage to FFT.

Phase structure (θ-space): the observed central-mass ratios (> 1) quantify that the time-aggregated phase law $\bar{\varphi}$ concentrates around $\theta = 0$, after ($m1$) anchoring. $GEM = 1.477$, $FFT = 1.437$, where values > 1 imply that the average density in $|\theta| \le \pi/2$ exceeds the global mean, which yields a unimodal, center-heavy phase law, consistent with the current DS settings after alignment. GEM is a bit more center-concentrated while FFT is a bit more spread out. This matches the slightly smoother FFT behavior reported by the CRPS and MLPD metrics.

After enforcing a common physical normalization in both the simulator and predictor (non-negative intensity channel before applying the Fraunhofer envelope), KPT-GEM and KPT-FFT yield consistent, high-quality reconstructions. The predicted screen-position densities $p(x)$ closely track the empirical distribution, with FFT exhibiting slightly better proper-scoring performance, ↑MLPD and ↓CRPS. The recovered time-aggregated phase laws $\bar{\varphi}$ concentrate around $\theta = 0$, central-mass ratios in $[1.44,\ 1.48$, while the aggregated kime-surface $\bar{S}$ shows the expected cosine-like contrast ($\bar{S}(0) > 0$, $\bar{S}(\pi) < 0$).

These results demonstrate that the KPT reconstructions are not just descriptive. Once mapped through the Fraunhofer envelope, they provide calibrated predictive densities for physical observables on the detection screen. This supports both forecasting and uncertainty quantification (UQ) in the double-slit setting. There is a small FFT advantage, which may be attributed to its spectral smoothing, which slightly improves predictive calibration without altering the physical interpretation.

Supporting website

The entire R markdown electronic notebook containing the end-to-end computational protocol and reproducible results are available on the supporting website. The complete pipeline workflow includes: This needs to be changed when we finalize the submission, the link inclusing the table of contents